US012412954B2

(12) United States Patent
Hernandez Saab et al.

(10) Patent No.: US 12,412,954 B2
(45) Date of Patent: Sep. 9, 2025

(54) **OPTIMIZED BATTERY ASSEMBLY VENTING**

(71) Applicant: **Rivian IP Holdings, LLC**, Plymouth, MI (US)

(72) Inventors: **Daniel Alfredo Hernandez Saab**, Irvine, CA (US); **Bruce Philip Edwards**, Menlo Park, CA (US); **Kyle Butterfield**, Ladera Ranch, CA (US); **Vignesh Sekar**, Westland, MI (US); **David John Revolinski**, Dana Point, CA (US)

(73) Assignee: **Rivian IP Holdings, LLC**, Plymouth, MI (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 561 days.

(21) Appl. No.: **17/890,025**

(22) Filed: **Aug. 17, 2022**

(65) **Prior Publication Data**

US 2023/0072437 A1 Mar. 9, 2023

Related U.S. Application Data

(60) Provisional application No. 63/242,290, filed on Sep. 9, 2021.

(51) **Int. Cl.**

H01M 50/325	(2021.01)
H01M 50/249	(2021.01)
H01M 50/291	(2021.01)

(52) **U.S. Cl.**

CPC ***H01M 50/325*** (2021.01); ***H01M 50/249*** (2021.01); ***H01M 50/291*** (2021.01); *H01M 2220/20* (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,557,415	B2	10/2013	Herron et al.	
8,557,416	B2	10/2013	Mardall et al.	
2007/0015046	A1	1/2007	Kim et al.	
2010/0273034	A1	10/2010	Hermann et al.	
2011/0174556	A1	7/2011	Hermann et al.	
2012/0231306	A1*	9/2012	Herron	H01M 50/342 429/53
2012/0237803	A1*	9/2012	Mardall	H01M 10/653 429/53
2020/0152926	A1	5/2020	Wynn et al.	

(Continued)

FOREIGN PATENT DOCUMENTS

CN 110190211 A * 8/2019

OTHER PUBLICATIONS

Machine Translation of CN 110190211A (Year: 2019).*

(Continued)

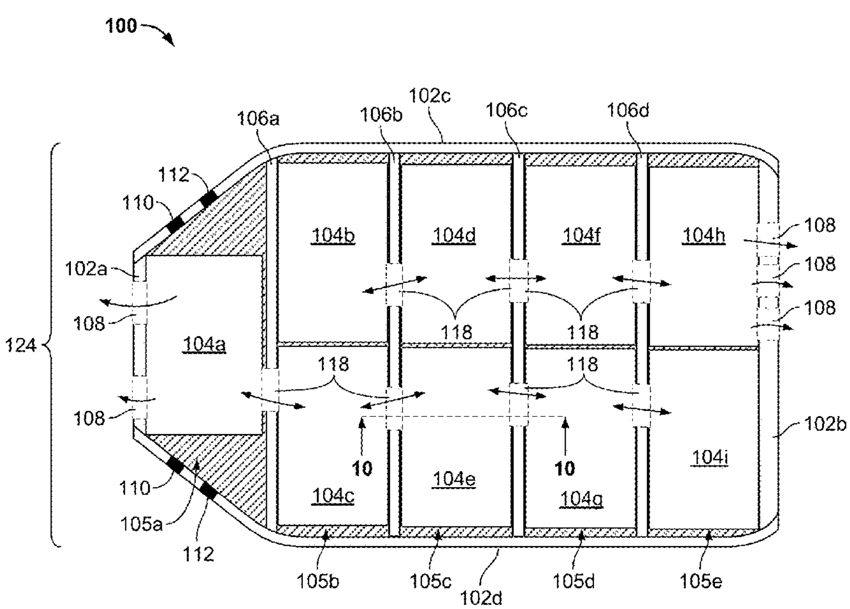
Primary Examiner — Gregg Cantelmo

(74) *Attorney, Agent, or Firm* — Haley Guiliano LLP

(57) **ABSTRACT**

Systems and methods are provided to vent a battery pack for an electric vehicle. The battery pack comprises a pack housing subdivided by a crossmember. The crossmember defines a multidirectional vent passage configured to allow venting with a first bay and a second bay. A pressure release valve is arranged on the pack housing and configured to release pressure based on the multidirectional vent passage.

20 Claims, 14 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

2020/0152935	A1*	5/2020	Wynn	.... B60L 50/64
2020/0152941	A1	5/2020	Wynn et al.	
2020/0365859	A1	11/2020	Hoefner	
2021/0184196	A1	6/2021	Dao et al.	
2022/0085452	A1	3/2022	Lee et al.	

OTHER PUBLICATIONS

U.S. Appl. No. 17/889,981, filed Aug. 17, 2022, Daniel Alfredo Hernandez Saab.
U.S. Appl. No. 17/889,984, filed Aug. 17, 2022, Daniel Alfredo Hernandez Saab.

* cited by examiner

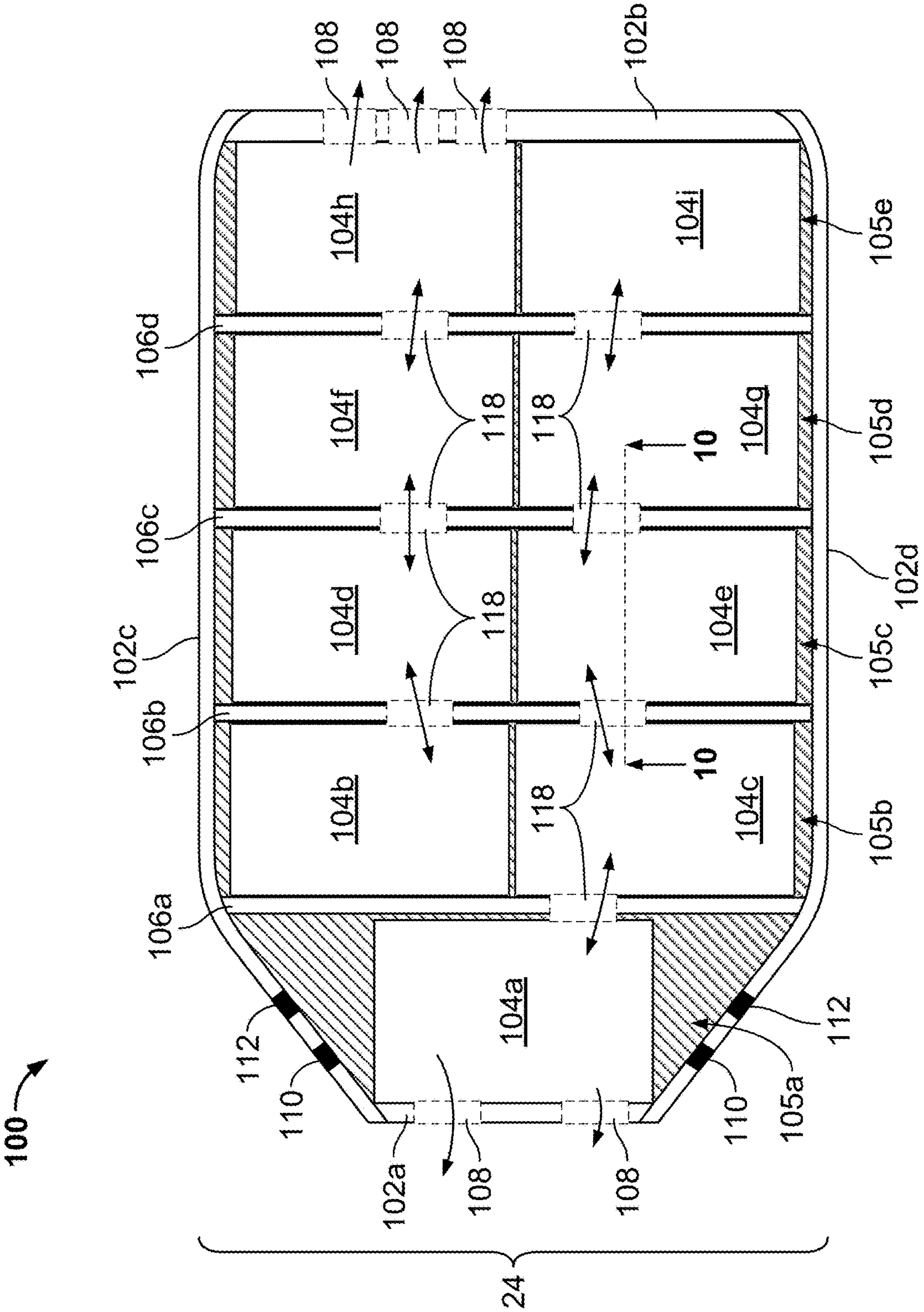

100
124
102a
102b
102c
102d
104a
104b
104c
104d
104e
104f
104g
104h
104i
105a
105b
105c
105d
105e
106a
106b
106c
106d
108
110
112
118
10

FIG. 1A

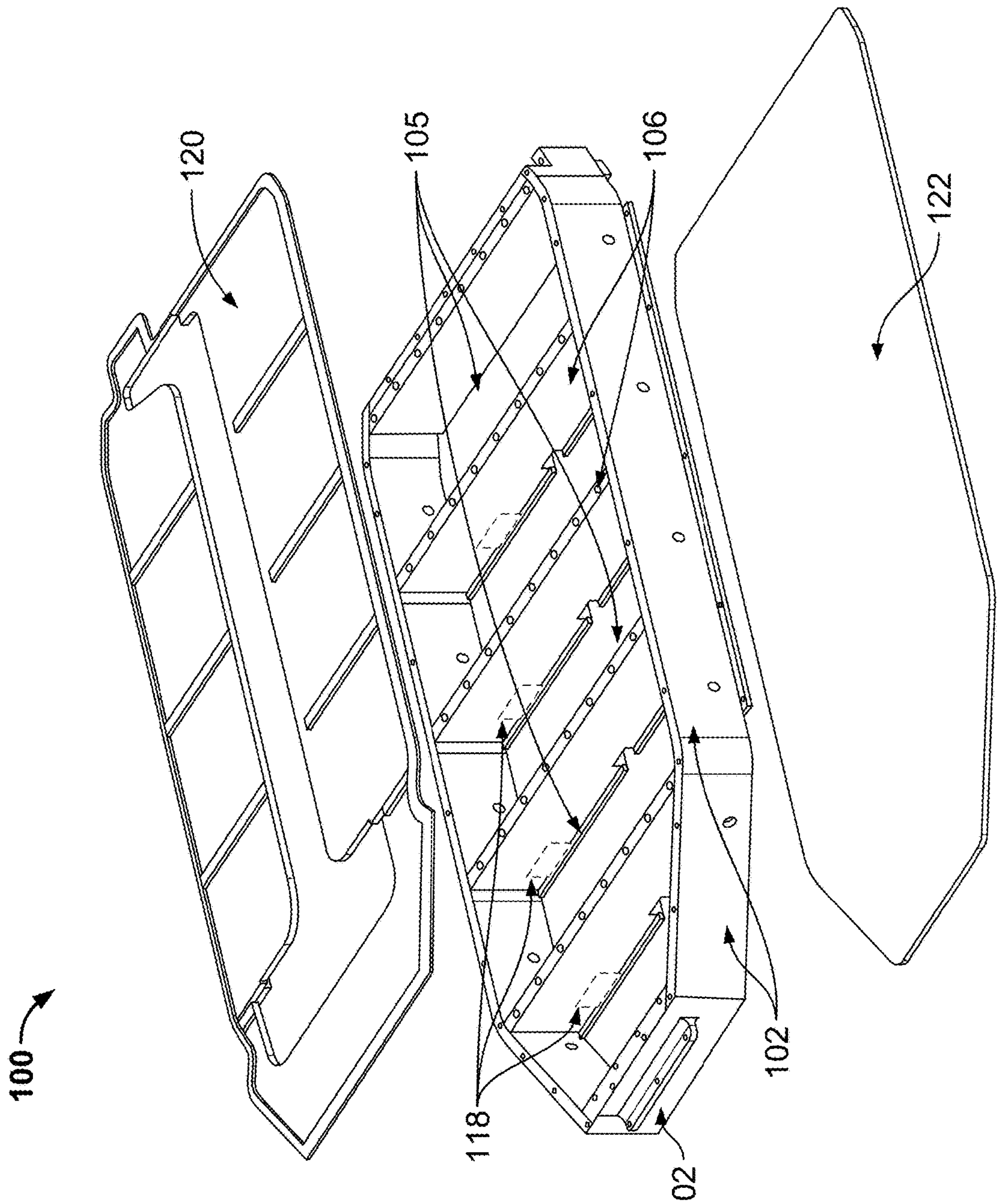


FIG. 1B

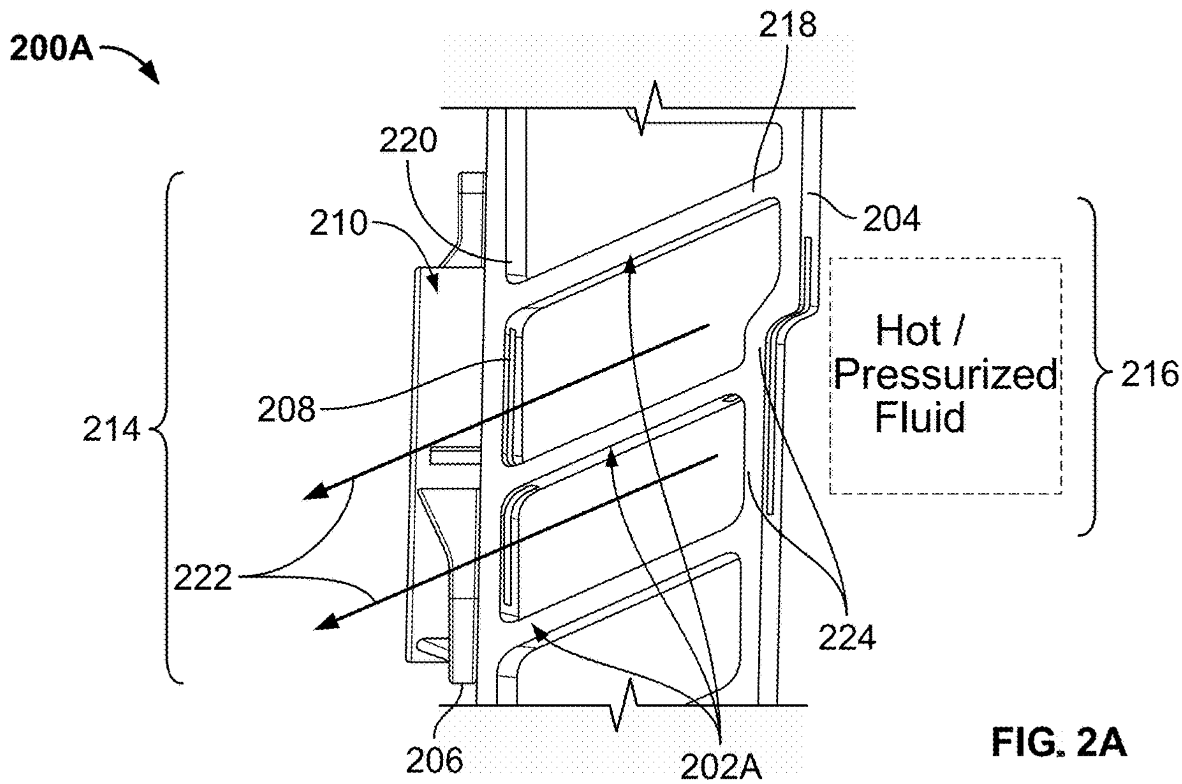

200A
218
220
210
204
Hot / Pressurized Fluid
216
208
214
222
224
206
202A

FIG. 2A

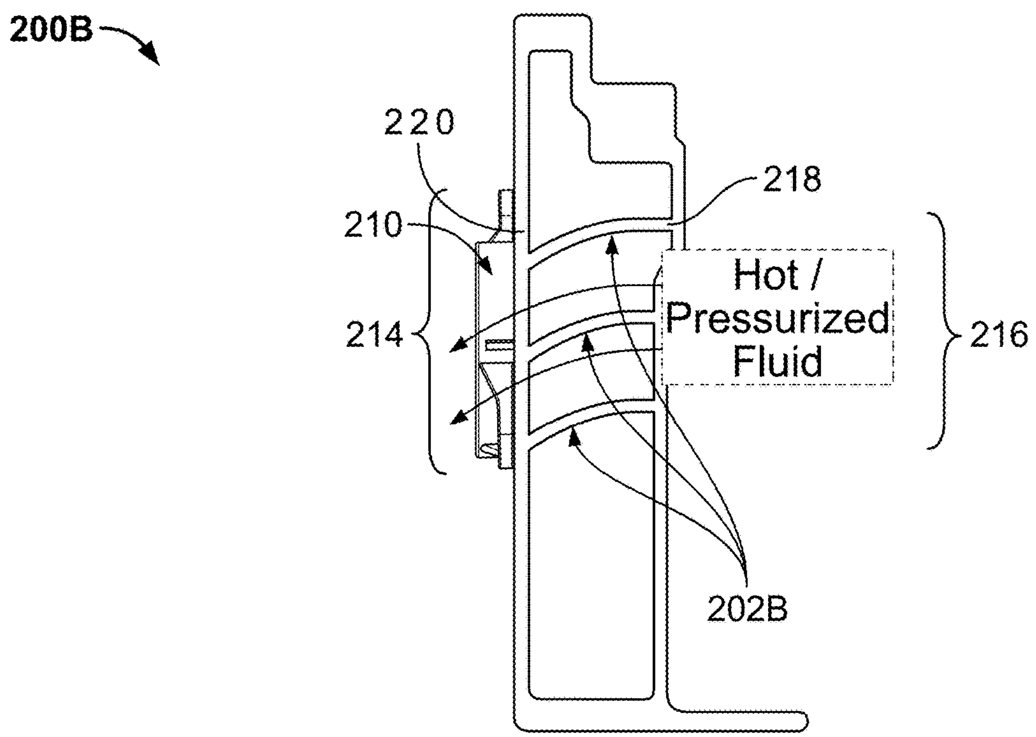

200B
220
218
210
Hot / Pressurized Fluid
214
216
202B

FIG. 2B

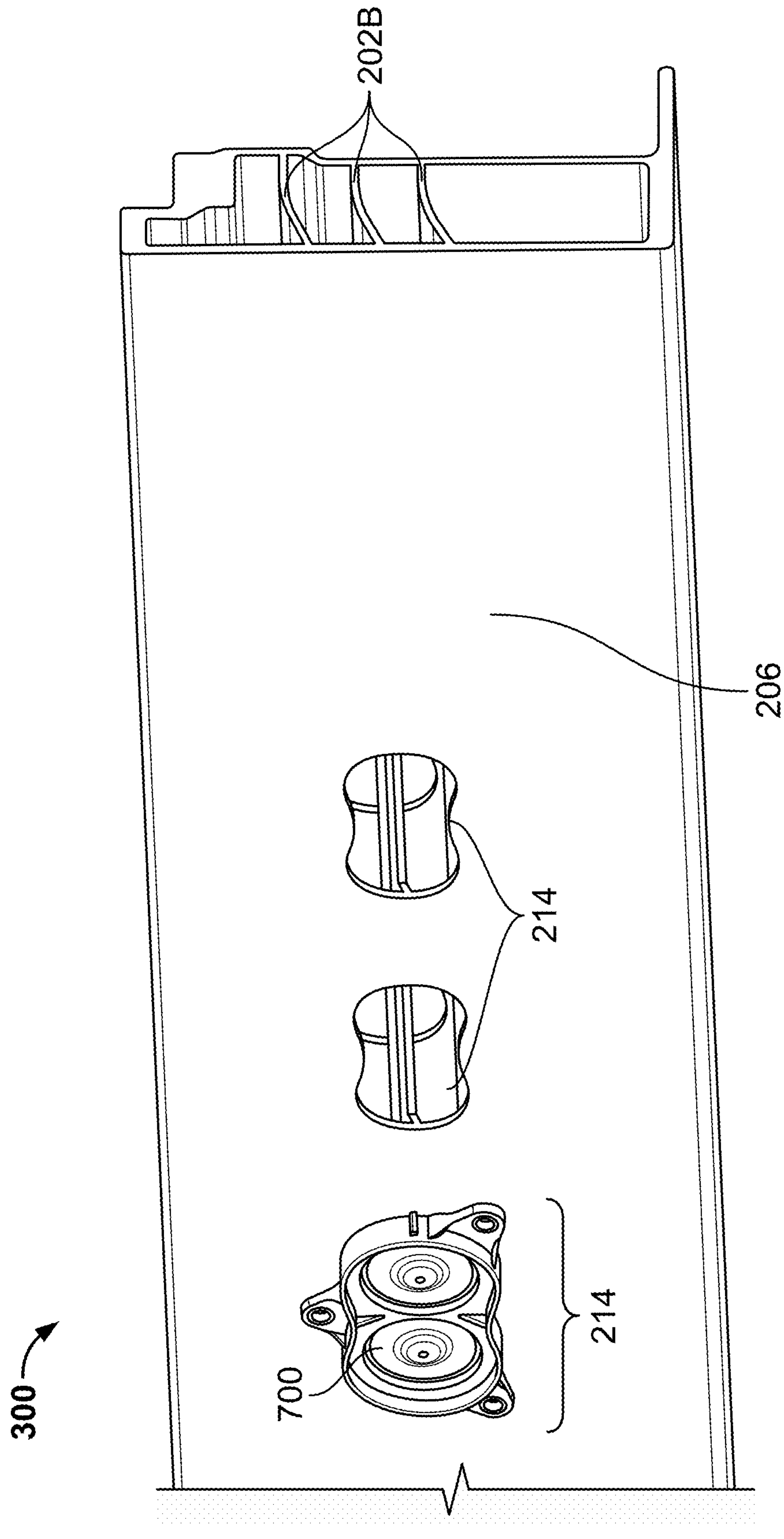


FIG. 3

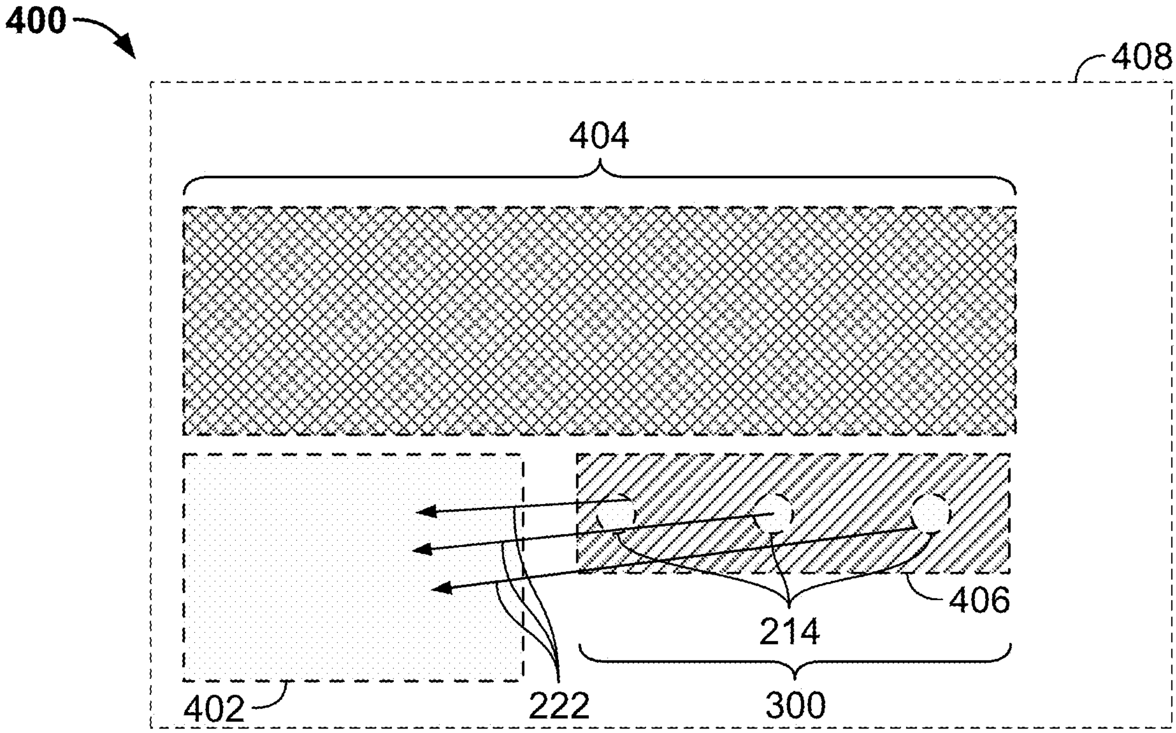

400
408
404
402
222
300
214
406

FIG. 4

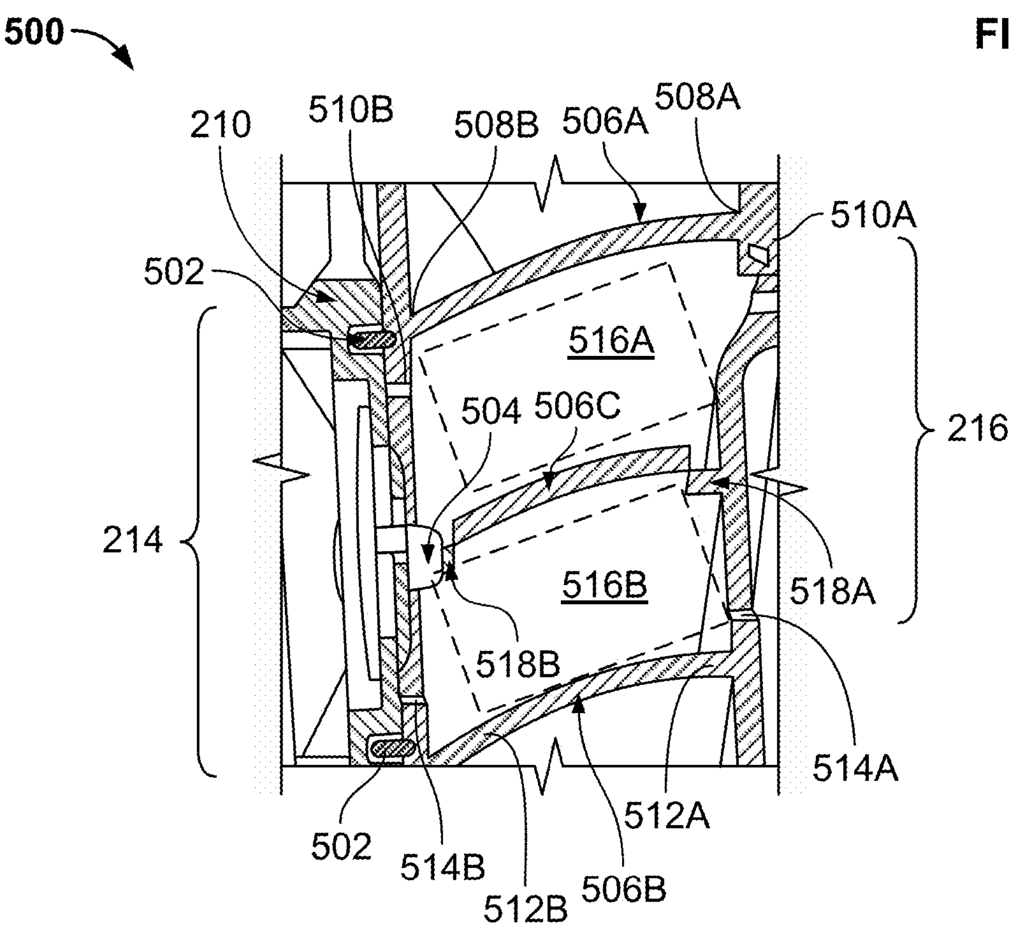

500
210
510B
508B
506A
508A
510A
502
516A
504
506C
216
214
518A
516B
518B
514A
502
514B
512B
506B
512A

FIG. 5

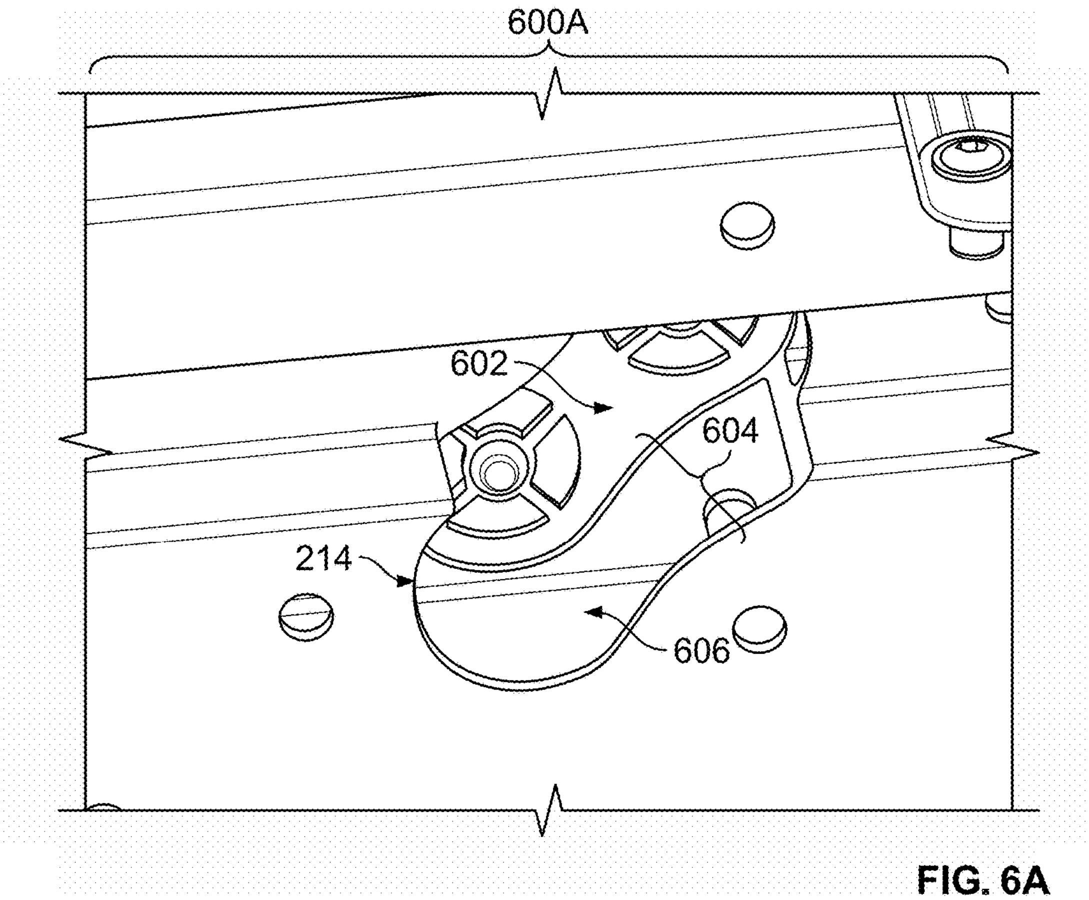


FIG. 6A

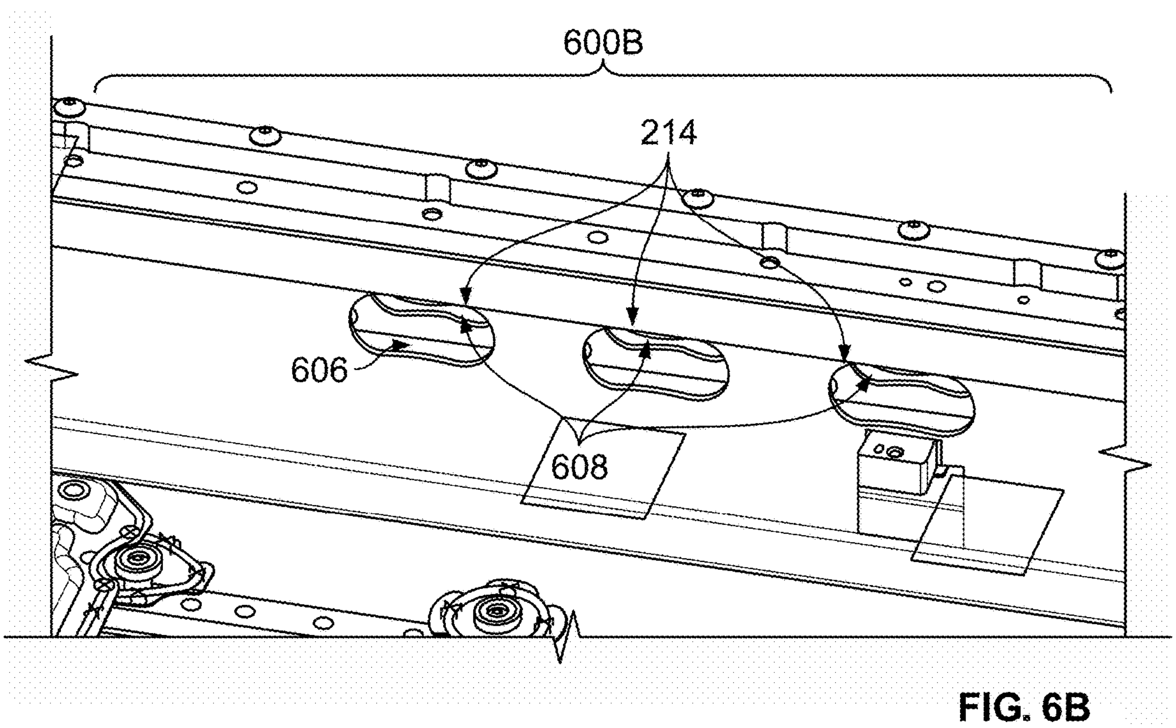


FIG. 6B

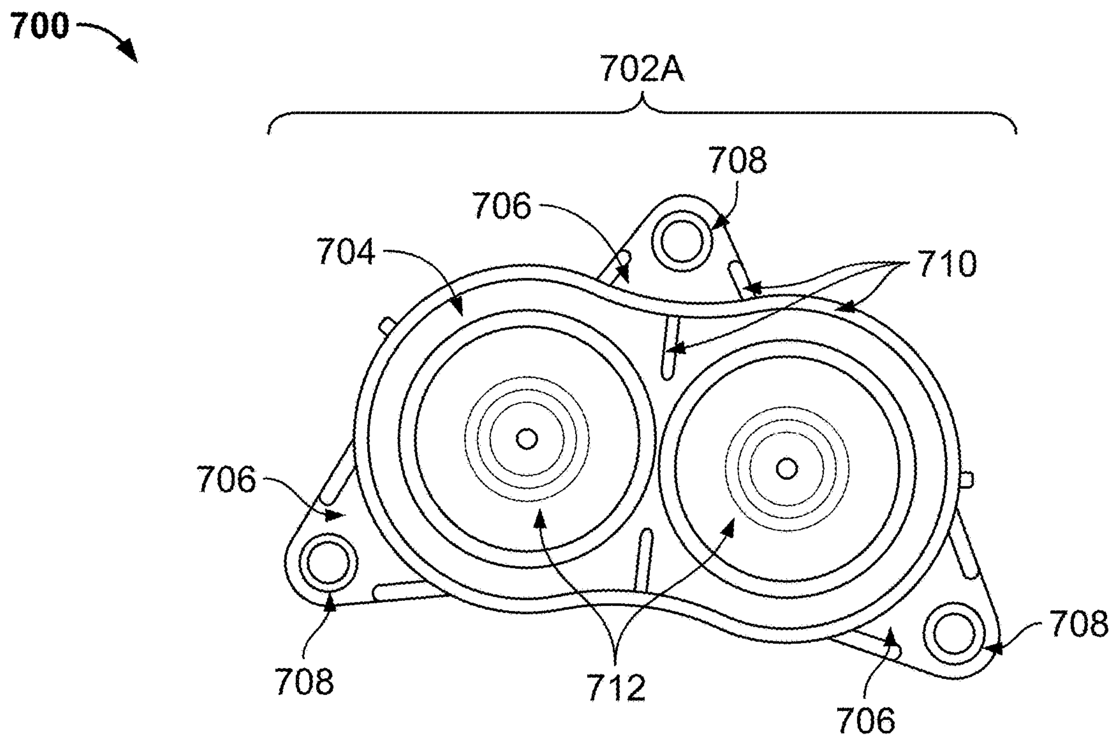

700
702A
708
706
710
704
706
712
708
708
706

FIG. 7A

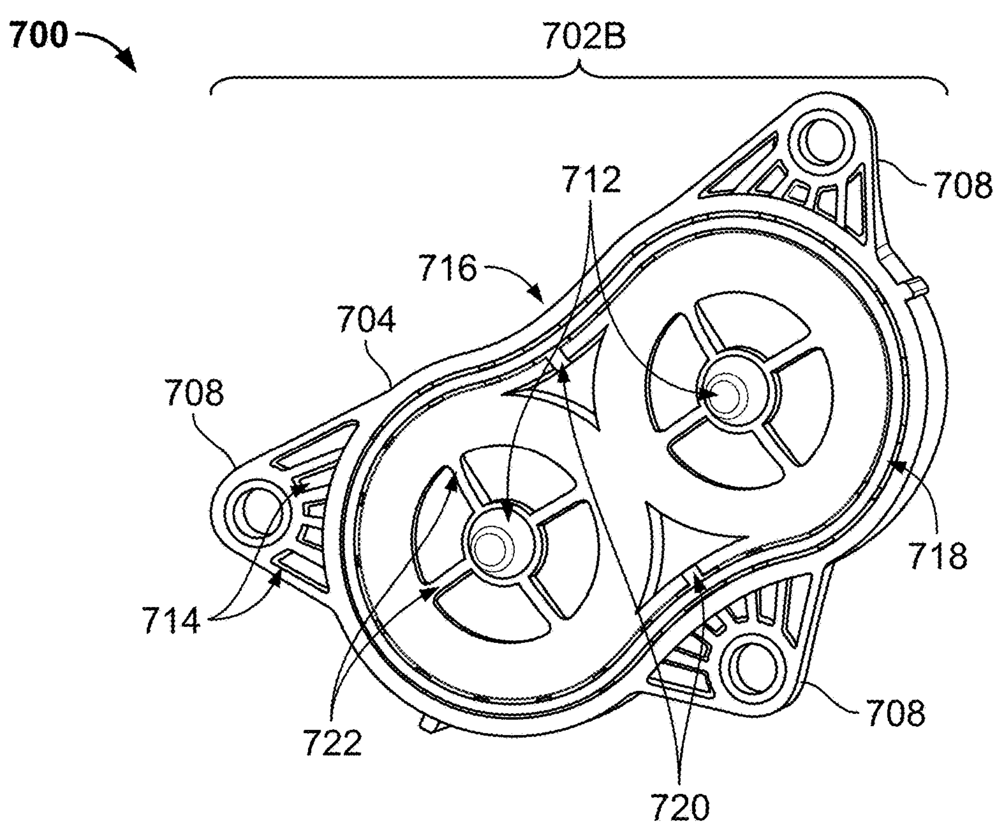

700
702B
708
712
716
704
708
718
714
722
720
708

FIG. 7B

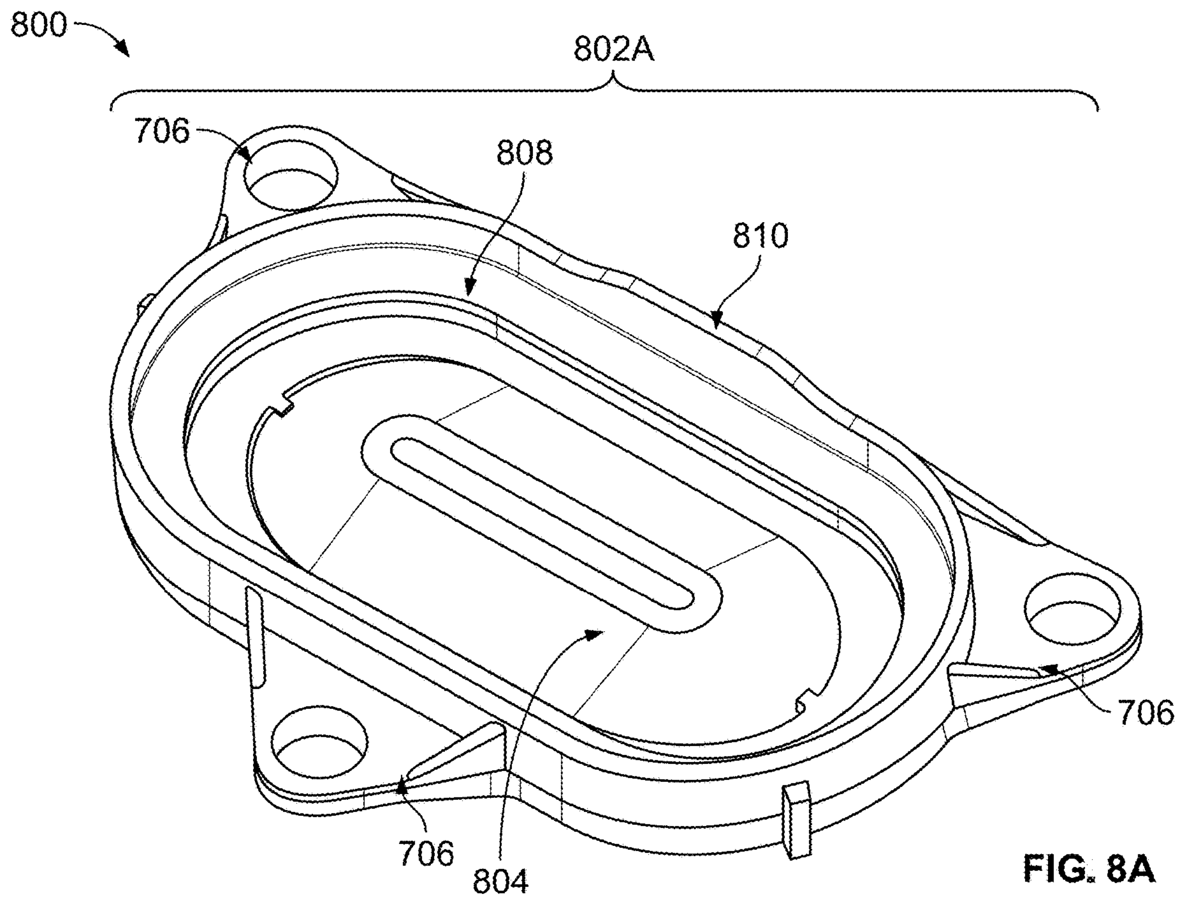

800
802A
706
808
810
706
706
804

FIG. 8A

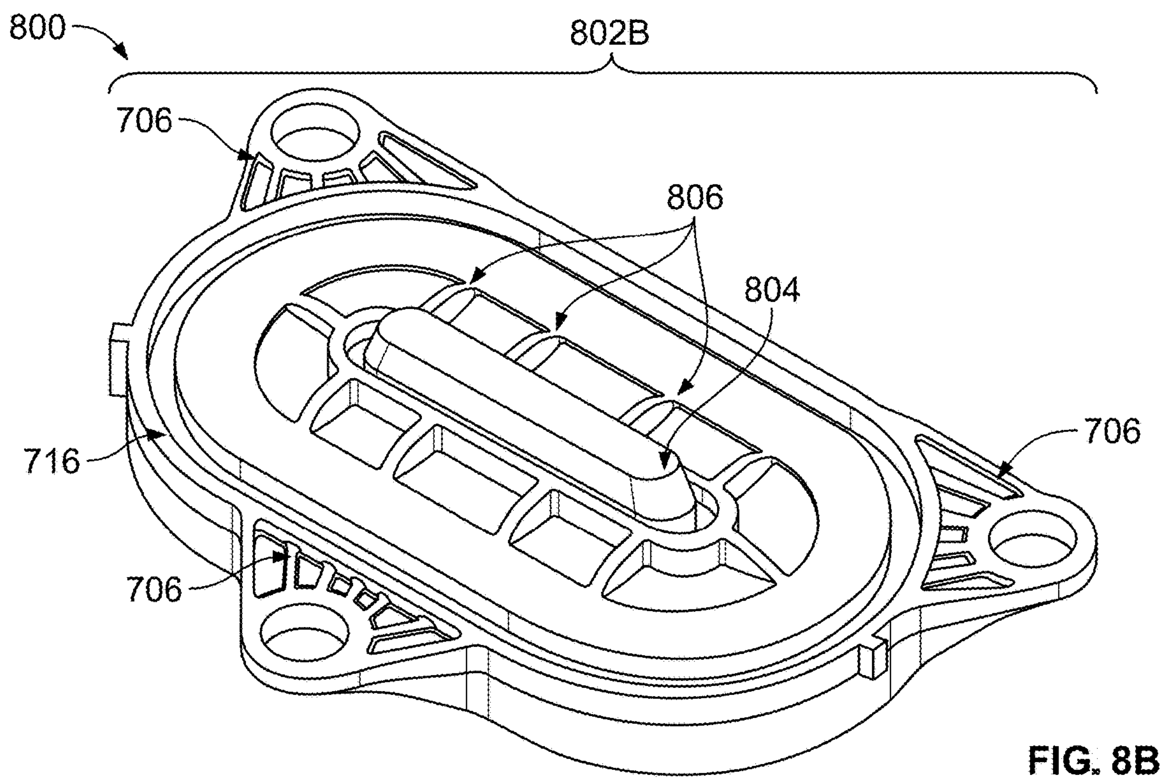

800
802B
706
806
804
716
706
706

FIG. 8B

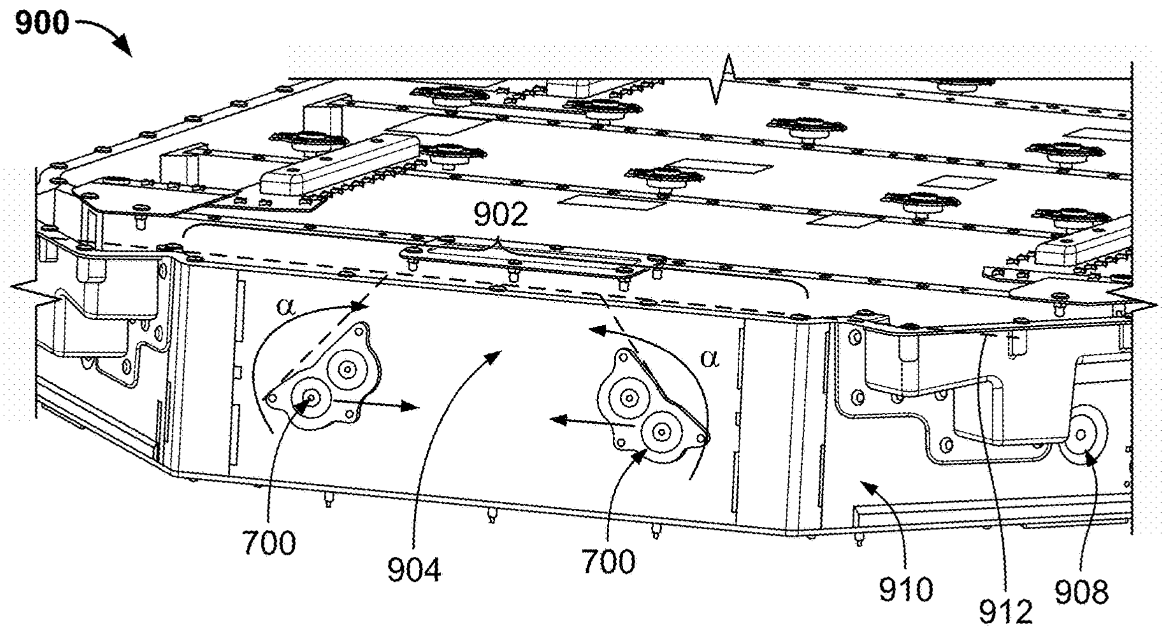


FIG. 9A

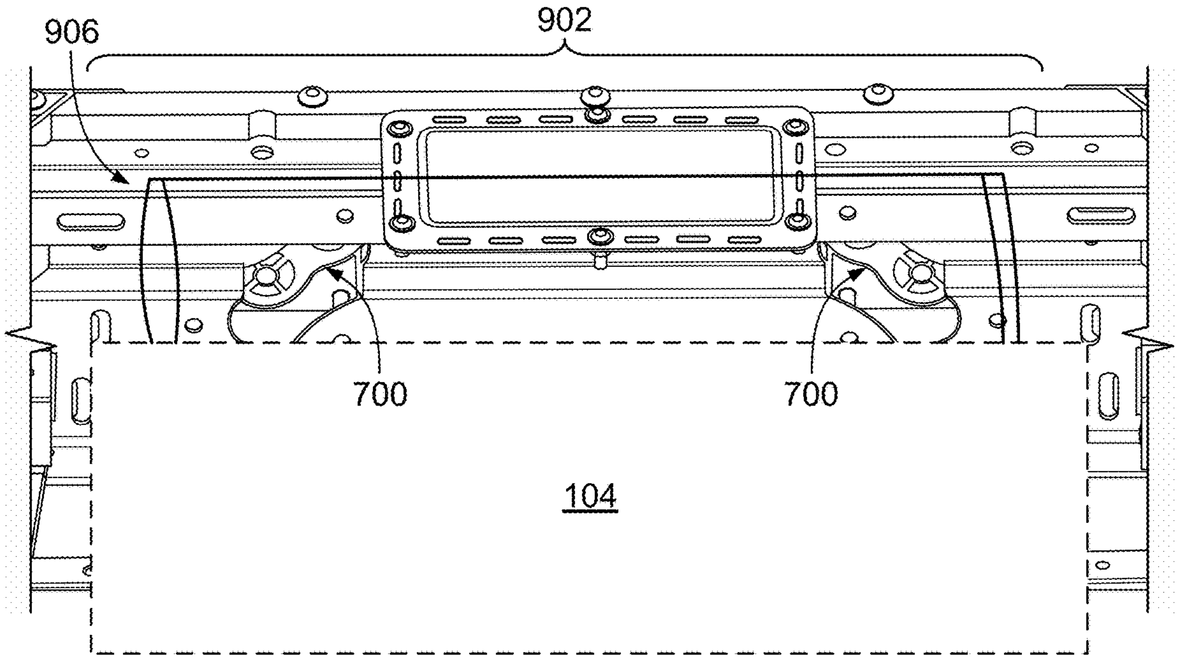


FIG. 9B

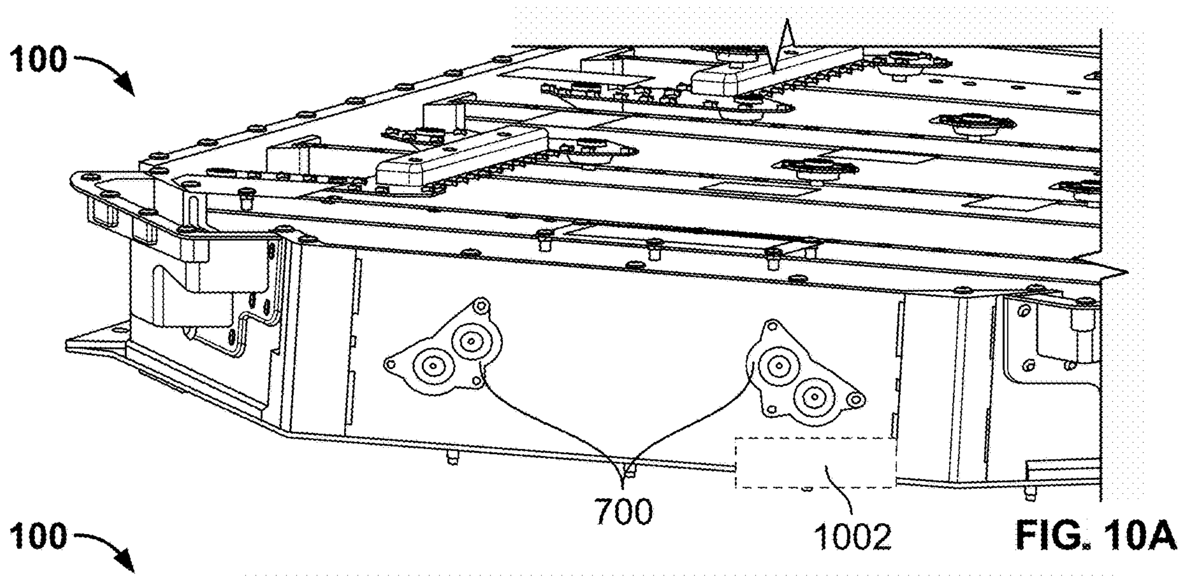


FIG. 10A

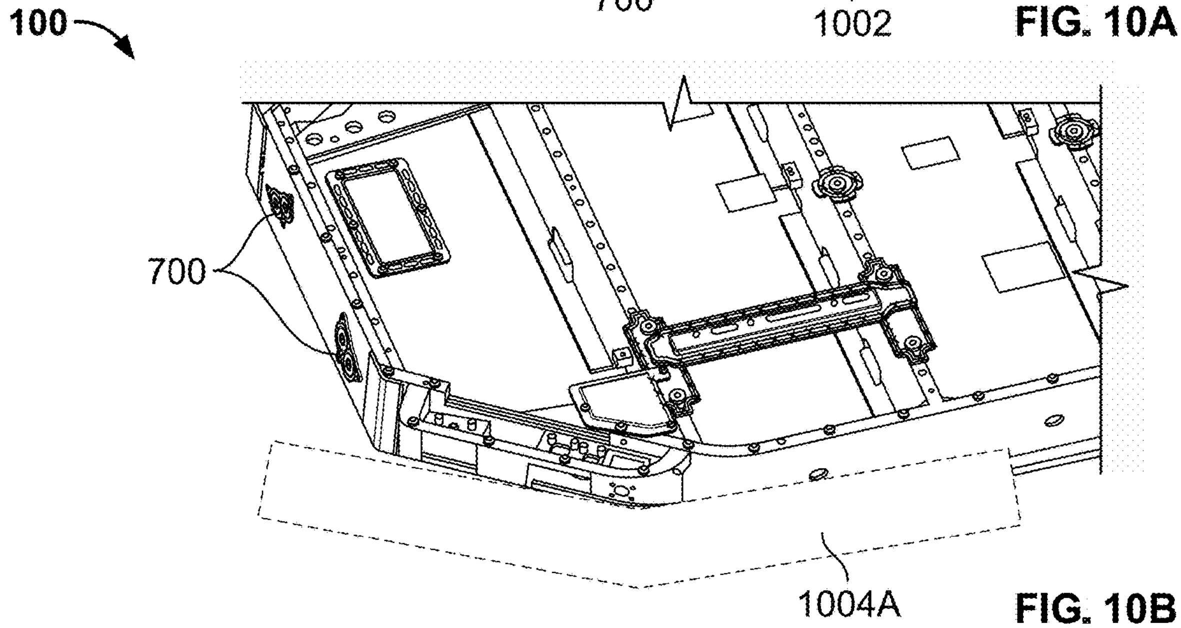


FIG. 10B

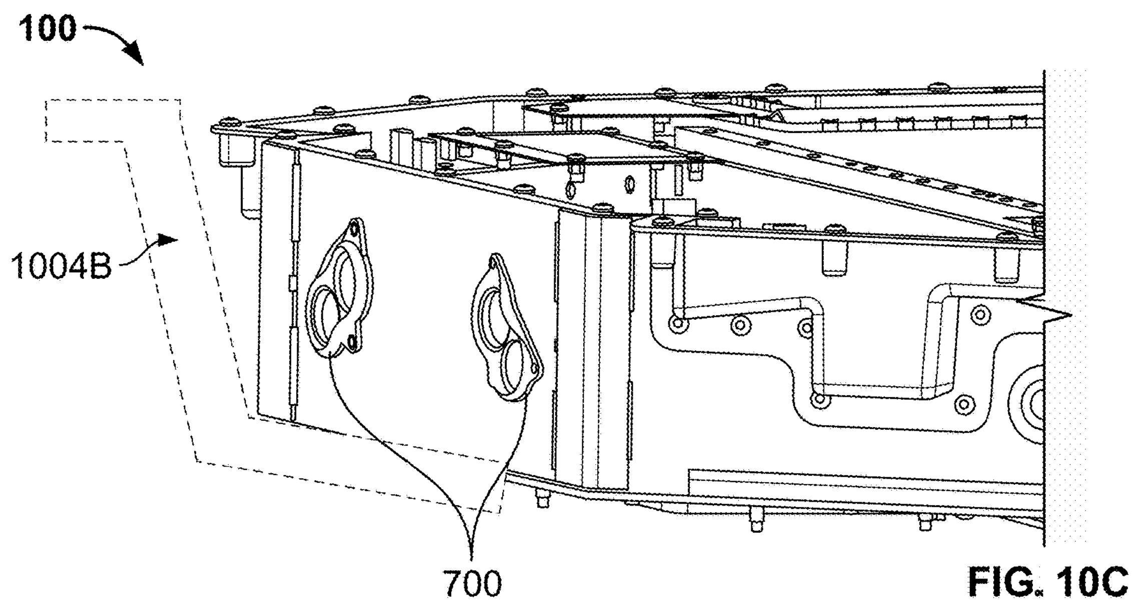


FIG. 10C

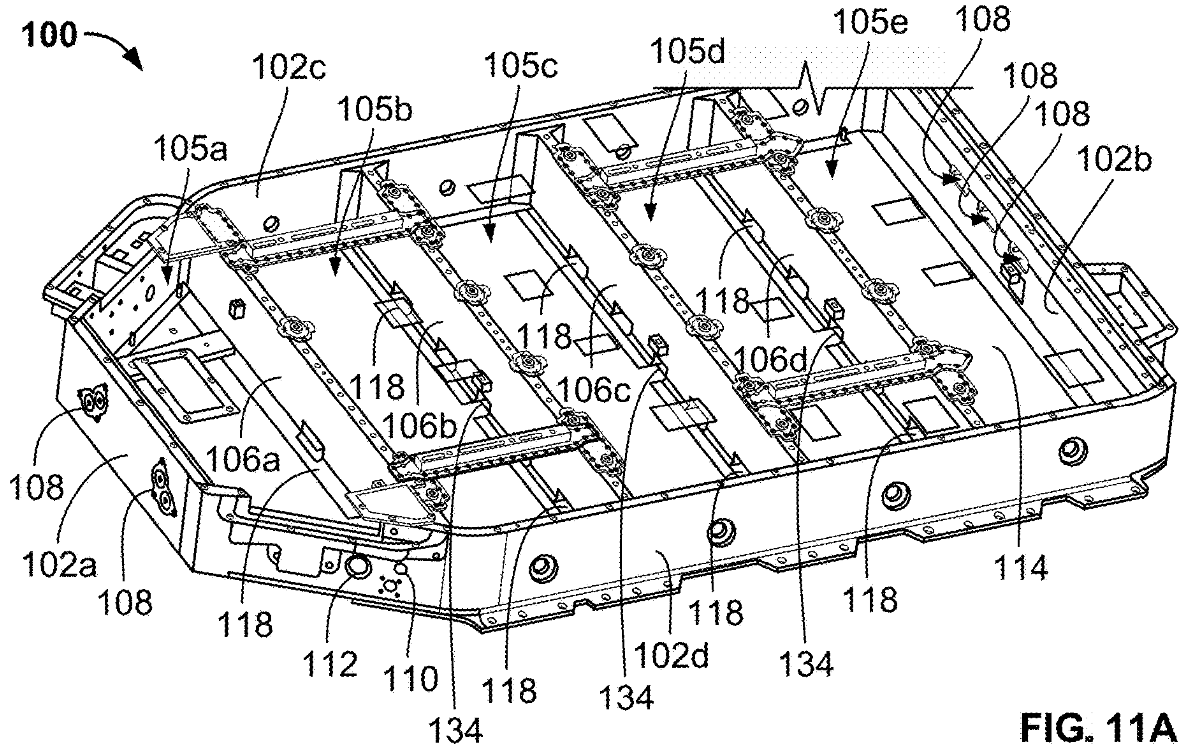


FIG. 11A

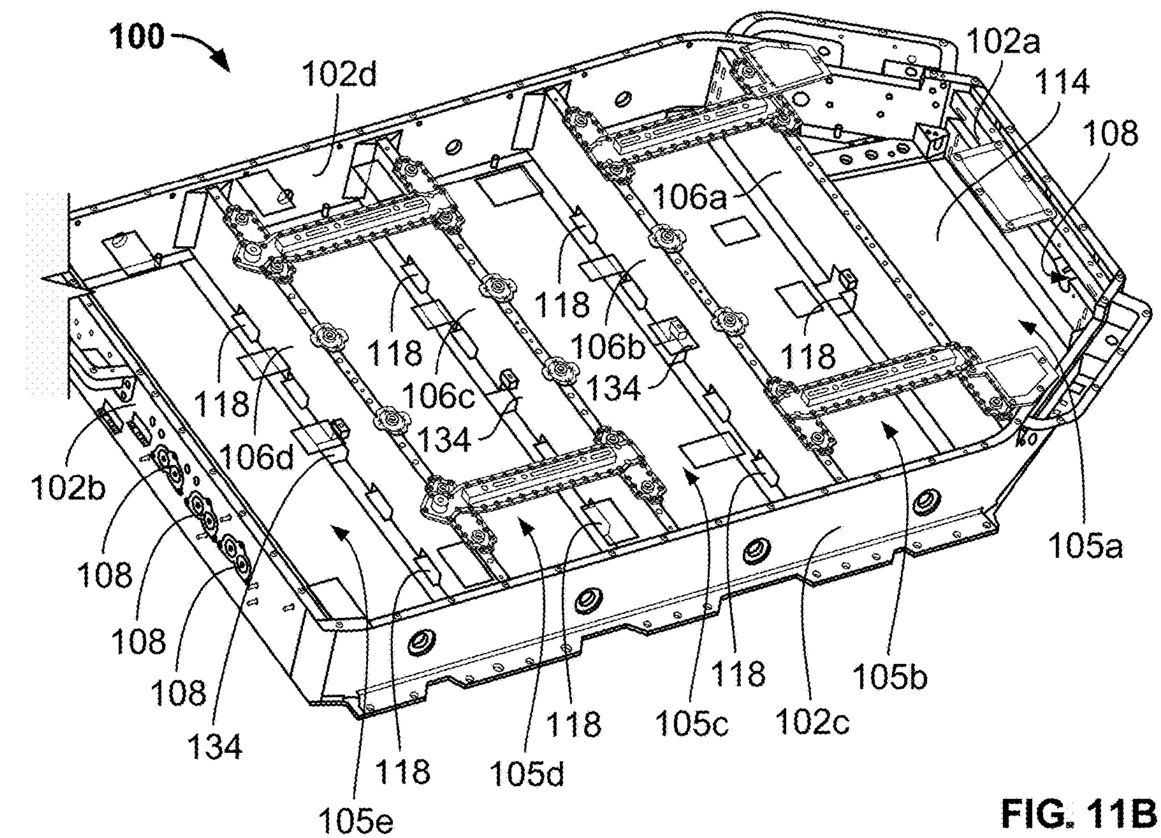


FIG. 11B

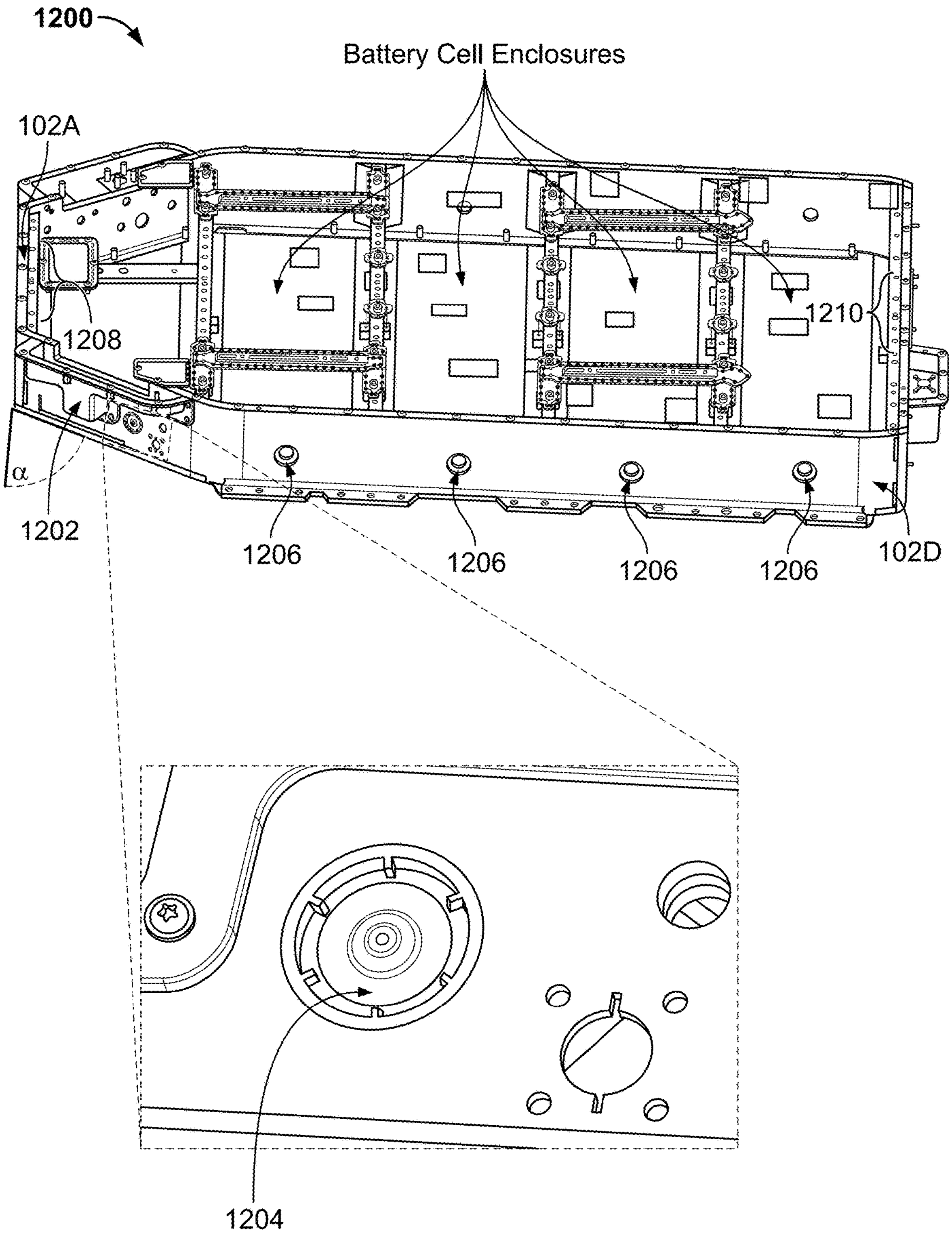

1200
Battery Cell Enclosures
102A
1208
1210
α
1202
1206
1206
1206
1206
102D
1204

FIG. 12

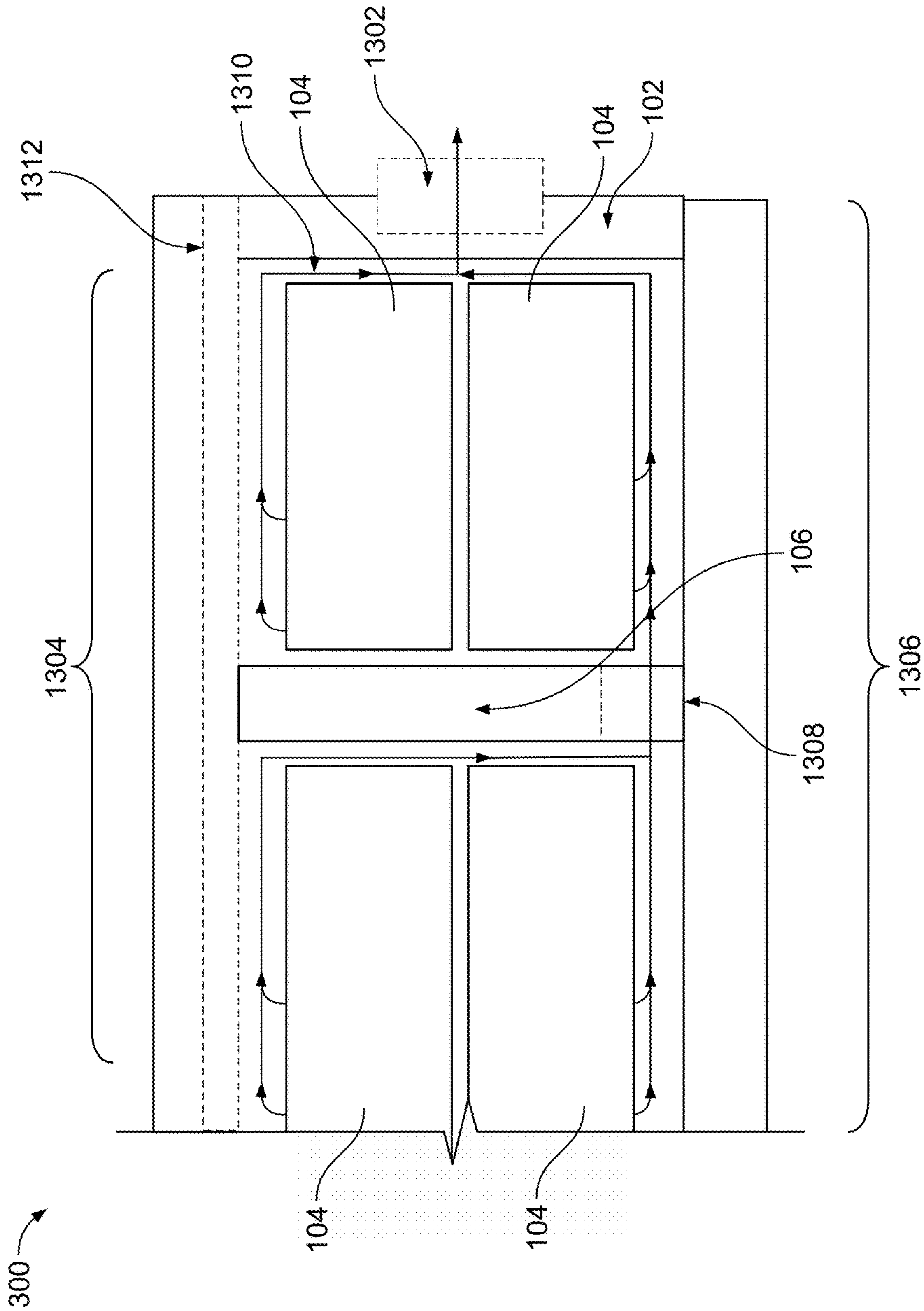


FIG. 13

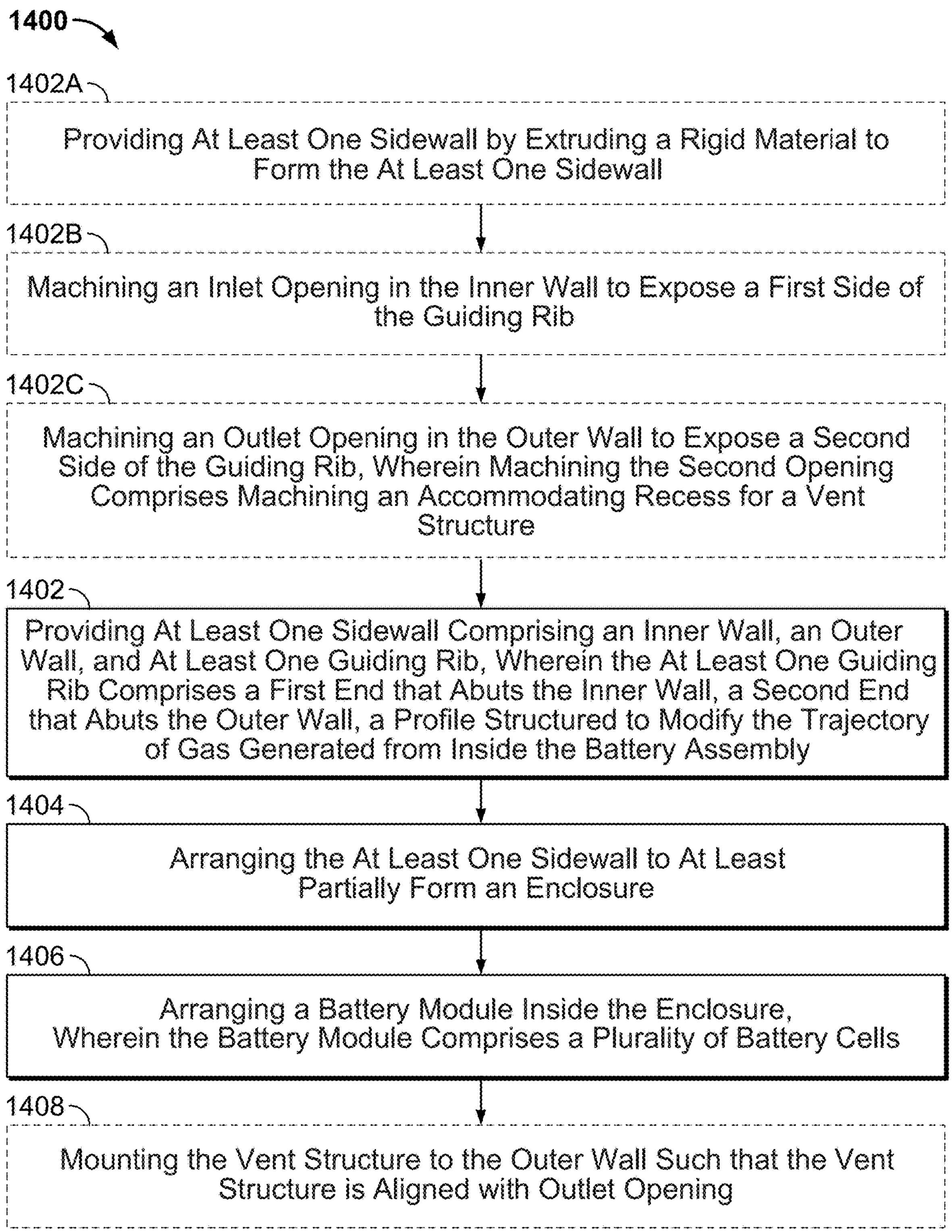

1400
1402A
Providing At Least One Sidewall by Extruding a Rigid Material to Form the At Least One Sidewall
1402B
Machining an Inlet Opening in the Inner Wall to Expose a First Side of the Guiding Rib
1402C
Machining an Outlet Opening in the Outer Wall to Expose a Second Side of the Guiding Rib, Wherein Machining the Second Opening Comprises Machining an Accommodating Recess for a Vent Structure
1402
Providing At Least One Sidewall Comprising an Inner Wall, an Outer Wall, and At Least One Guiding Rib, Wherein the At Least One Guiding Rib Comprises a First End that Abuts the Inner Wall, a Second End that Abuts the Outer Wall, a Profile Structured to Modify the Trajectory of Gas Generated from Inside the Battery Assembly
1404
Arranging the At Least One Sidewall to At Least Partially Form an Enclosure
1406
Arranging a Battery Module Inside the Enclosure, Wherein the Battery Module Comprises a Plurality of Battery Cells
1408
Mounting the Vent Structure to the Outer Wall Such that the Vent Structure is Aligned with Outlet Opening

FIG. 14

OPTIMIZED BATTERY ASSEMBLY VENTING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Patent Application No. 63/242,290 filed Sep. 9, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.

INTRODUCTION

The present disclosure is directed to optimized vents and venting of battery assemblies. The present disclosure is also directed to directing flows of heat or pressure out of battery assemblies to avoid a disruption in the operation of the battery assemblies.

SUMMARY

The present disclosure is directed to systems and methods for venting heat and pressure generated by battery cells and other conditions within a battery assembly, and more particularly, to systems and methods that utilize strategically shaped and positioned pressure release valves and other structural features embedded in the walls of the battery assembly to optimize venting of heat and pressure such that the heat and pressure egress the battery assembly. A battery assembly or battery pack incorporating these systems and methods is advantageous in that the pressure release valves, and other structural features, improve efficiency of the operation of the battery cells, reduces environmental stress caused by pressurized gas, and improves overall packaging of the battery assembly as the battery assembly is configured to accommodate the components installed in the environment surrounding the battery assembly.

The battery assembly may comprise at least one sidewall arranged to encompass in part a plurality of battery cells. In some embodiments, the at least one sidewall may comprise an inner surface comprising an inlet opening configured to receive gas generated by the plurality of battery cells, an outer surface comprising an outlet opening configured to vent the gas, and at least one guiding rib that modifies a trajectory of gas (e.g., at least one of ambient air or pressured gas generated as a result of heated battery cells changing conditions of the ambient air in a battery assembly) between the inner surface and the outer surface. The sidewall is advantageous in that it provides an avenue by which pressurized gas can be vented out of the battery assembly and the profile of the guiding rib between the inlet and the outlet stabilize the flow of the pressurized gas and direct the pressurized gas away from at least one of surrounding components or areas that may be occupied by vehicle occupants.

The battery assembly may additionally incorporate at least one pressure release valve, which may be mounted on at least one of the previously described sidewalls which may be referred to as mounting walls herein. The position of the pressure release valve once installed in the sidewall or mounted to the mounting wall may correspond to a position of the previously mentioned outlets in the outer surface. In order to enable optimal venting, the pressure release valve may comprise a housing configured to be secured to a mounting wall, a first membrane positioned towards a first side of the housing, a second membrane positioned adjacent to the first membrane and positioned towards a second side of the housing, and a gasket structured to form a continuous seal between the housing and the mounting wall. The membranes and gasket may be formed out of any material suitable for the temperatures corresponding to an operational battery assembly such that the material is able to maintain a watertight seal once the pressure release valve is installed. In some embodiments, the membranes may comprise at least one of a moveable cover, or a rotating enclosure surface. In some embodiments, portions of the housing may be structured to mechanically deform (e.g., melt such that at least one membrane falls out of the housing) when subjected to the conditions corresponding to thermal runaway resulting from the operation of the battery cells in the battery assembly. When the portions of the housing mechanically deform, the membranes may fall out of the pressure release valve housing in response to being exposed to the thermal runaway condition and create an opening in the mounting wall to enable rapid egress of pressurized gas from within the mounting wall, which may be guided by the previously mentioned guiding rib through the mounting wall to the pressure release valve housing.

The battery assembly may be configured for installation and use in a vehicle. The walls of the battery assembly may comprise the previously discussed sidewall and may comprise a forward-facing wall configured to be positioned towards a front of the vehicle, a rear-facing wall configured to be positioned towards a rear of the vehicle, a first plurality of pressure release valves embedded in the forward-facing wall, wherein the first plurality of pressure release valves is arranged to enable the egress of pressurized gas from between the forward-facing and rear-facing walls, and a second plurality of pressure release valves embedded in the second wall, wherein the second plurality of pressure release valves are arranged to enable the egress of gas from between the forward-facing and rear-facing walls. Each of the first and second plurality of pressure release valves may comprise the two membrane pressure release valves previously discussed. In some embodiments, the two membrane pressure release valves are paired with single membrane pressure release valves to achieve a target egress rate of the pressurized gas generated by the operation of the battery cells within the battery assembly. In some embodiments, there may be a pair of angled two membrane pressure release valves on the forward-facing wall with a set of three two membrane pressure release valves embedded in the rear-facing wall and positioned level with a flat plane defined by the top or the bottom of the rear-facing wall, depending on a determined location of an optimal outlet to ensure a maximum target egress rate. For example, each battery assembly may have a variety of wall positions and features which may affect the path of pressurized gas throughout the assembly. An optimal outlet position for either the first or second plurality of pressure release valves may balance target stiffness of the wall to survive crash conditions, maximum water ingress allowable for the battery assembly, and pressurized gas paths within the battery assembly.

In some embodiments, a battery pack, e.g., for an electric vehicle, includes a pack housing having multiple module bays, e.g., with a crossmember subdividing an interior of the pack housing into two adjacent module bays. One or both of the module bays may be configured to support or provide storage for one or more batteries or battery modules. The crossmember defines a bidirectional or multidirectional vent passage configured to allow fluid communication between the adjacent module bays. The battery pack may also include a pressure release valve positioned at a perimeter of the pack housing. Accordingly, excess pressure from a module bay may pass through the bidirectional vent passage to the one-way vent for venting out of the battery pack. In some embodiments the crossmember that is arranged with the sidewall to form a first enclosure separated from a second enclosure by the crossmember. The crossmember comprises a guiding rib that modifies a trajectory of gas that transfers from the second enclosure to the first enclosure.

In some embodiments, the vent passage may be a slot or window in a crossmember, wall, or other structure between the modules or bays. Gas emitted from a battery cell in one module may thereby flow through the bidirectional vent passage to another module, with the gas being directed out of the adjacent module from the battery pack (e.g., along, or out of a perimeter of the battery pack). The specific locations of the slots, size of the slots, locations and number of vents or pressure release valves from the battery pack may be determined to balance structural support and necessary venting pressure based upon expected pressure/heat generation by the battery cells.

Bidirectional or multidirectional vent passage(s), in at least some embodiments, may freely allow fluid communication between modules, thereby facilitating relatively free flow of gas discharged from a battery cell along a desired flow path out of the battery pack. In some embodiments, the bidirectional vent passage is open and unobstructed by wiring or other components of the battery pack to provide a direct path for fluid communication between the adjacent module bays. The bidirectional vent passage(s) may also facilitate equalization of pressure between the module bays. As a result, gas that is discharged from a cell in one module or bay of the battery pack may be routed effectively to a pressure release valve or one-way pressure release valve which provides a vent path out of the battery pack. Vent flow paths from the modules may be provided to one or more desired locations on the battery pack and may allow consolidation of flow paths for discharged gas out of the battery pack, as will be discussed further below.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure, in accordance with one or more various embodiments, is described in detail with reference to the following figures. The drawings are provided for purposes of illustration only and merely depict typical or example embodiments. These drawings are provided to facilitate an understanding of the concepts disclosed herein and should not be considered limiting of the breadth, scope, or applicability of these concepts. It should be noted that for clarity and ease of illustration, these drawings are not necessarily made to scale.

FIG. **1**A depicts a top view of an exemplary battery pack, in accordance with some embodiments of the disclosure;

FIG. **1**B depicts an exploded view of an exemplary battery pack, in accordance with some embodiments of the disclosure;

FIGS. **2**A and **2**B each depict a respective cross sectional view of a battery pack sidewall comprising multiple guiding ribs, in accordance with some embodiments of the disclosure;

FIG. **3** depicts an outer surface of an exemplary battery pack sidewall with an embedded pressure release value, in accordance with some embodiments of the disclosure;

FIG. **4** depicts a side view of an exemplary battery assembly comprising at least one vented sidewall arranged to enable rapid egress of at least one of pressurized or hat gas generated within the battery assembly, in accordance with some embodiments of the disclosure;

FIG. **5** depicts a cross sectional view of an exemplary sidewall comprising guiding ribs, in accordance with some embodiments of the disclosure;

FIG. **6**A depicts an example of a battery pack sidewall, with a pressure release valve embedded in the battery pack sidewall at an angle, comprising an opening configured to collect moisture, in accordance with some embodiments of the disclosure;

FIG. **6**B depicts an example of a battery pack sidewall, with multiple pressure release valves embedded in the battery pack sidewall, in accordance with some embodiments of the disclosure;

FIG. **7**A depicts a top view of an exemplary dual membrane pressure release valve, in accordance with some embodiments of the disclosure;

FIG. **7**B depicts a bottom view of an exemplary dual membrane pressure release valve, in accordance with some embodiments of the disclosure;

FIG. **8**A depicts a top view of an exemplary single membrane pressure release valve, in accordance with some embodiments of the disclosure;

FIG. **8**B depicts a bottom view of an exemplary single membrane pressure release valve, in accordance with some embodiments of the disclosure;

FIGS. **9**A and **9**B each depict a respective view of an exemplary battery pack sidewall with a pair of angled dual membrane pressure release valves in accordance with some embodiments of the disclosure;

FIGS. **10**A, **10**B, and **10**C each depict a respective view of an exemplary battery assembly with a plurality of pressure release valves, in accordance with some embodiments of the disclosure;

FIGS. **11**A and **11**B each depict a respective top view of an exemplary battery assembly with a plurality of battery module bays, in accordance with some embodiments of the disclosure;

FIG. **12** depicts an exemplary battery assembly comprised of a plurality of battery module bays and a plurality of pressure release valves, in accordance with some embodiments of the disclosure;

FIG. **13** depicts a top view of an exemplary battery assembly with flow paths indicative of intended flow of at least one of pressurized or hot gas generated based on the operation of the exemplary battery modules, in accordance with some embodiments of the disclosure; and

FIG. **14** is a flowchart depicting an exemplary method of optimizing venting of at least one of pressurized or heated gas generated within a battery assembly, accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

Methods and systems are provided herein for venting heat and pressure generated by battery cells and other conditions within a battery assembly, and more particularly, to systems and methods that utilize strategically shaped and positioned pressure release valves and other structural features embedded in the walls of the battery assembly to optimize venting of heat and pressure such that the heat and pressure egress the battery assembly.

FIG. **1**A depicts a top view of battery pack **100** (e.g., a battery assembly, a battery pack assembly, a battery pack system, or any combination thereof). FIG. **1**B depicts an exploded view of battery pack **100** with battery pack cover **120** and battery pack base **122**, in accordance with some embodiments of the disclosure. Battery pack **100** may comprise more or fewer than the components or features depicted in FIGS. **1**A and **1**B. Battery pack **100** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **2**A-**13**. Additionally, battery pack **100** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

As shown in FIG. **1**A, battery pack **100** represents an example embodiment of the systems and methods described herein. Battery pack **100** is enclosed by sidewalls **102***a*, **102***b*, **102***c*, and **102***d* (collectively, **102**) extending about a perimeter of battery pack **100**. Battery pack **100** encloses battery modules **104***a*, **104***b*, **104***c*, **104***d*, **104***e*, **104***f*, **104***g*, **104***h*, and **104***i* (collectively, **104**), which collectively provide power to an apparatus or system requiring electrical power for operation (e.g., an electric vehicle powertrain system configured to enable movement of the electric vehicle). Battery modules **104** are enclosed within module bays **105***a*, **105***b*, **105***c*, **105***d*, and **105***e* (collectively, **105**), which are defined by sidewalls **102** and crossmembers **106***a*, **106***b*, **106***c*, and **106***d* (collectively, **106**). Collectively, sidewalls **102** and crossmembers **106** form housing **124**. Each bay of bays **105** is configured to accommodate at least one of battery modules **104**, and as shown in FIG. **1**A bay **105***a* is configured to accommodate a single one of battery modules **104** and the each of the remaining bays **105***b-e* is configured to accommodate two of battery modules **104**. In some embodiments, any number of modules **104** may be positioned in any of bays **105** (e.g., depending on the dimensions of at least one of housing **124**, any of bays **105**, or battery modules **104**). Crossmembers **106** each extend laterally along battery pack **100** between sidewalls **102***c* and **102***d*.

Battery pack **100** may be substantially sealed such that fluid flow (e.g., at least one of pressurized or heated gas into and out of the enclosure defined by sidewalls **102**) is limited to specific venting flow paths. Venting flow paths, as shown in FIG. **1**A, are provided by pressure release valves **108**, plug valves **110**, and deformable pressure release valves **112**. Pressure release valves **108**, **110**, and **112** are generally configured to handle different levels of pressure/flow from battery pack **100** to an external environment surrounding housing **124** of battery pack **100**. In some embodiments, plug valves **110** may facilitate relatively low-pressure flows to and from battery pack **100**, which may occur due to at least one of thermal expansion or contraction of gas within battery pack **100** corresponding to the operation of battery modules **104**. Plug valves **110** may be a generally solid plug of permeable material configured to permit a maximum level of flow that is relatively low, consistent with thermal expansion/contraction of battery pack **100**.

Pressure release valves **108** are configured to facilitate one-way flow out of battery pack **100** in response to relatively greater thermal or pressure flows than ambient conditions within battery pack **100** prior to operation of battery modules **104** within module bays **105**. Pressure release valves **108** may each include moveable pressure release valve members, such as an umbrella valve or umbrella membrane. In some embodiments, the membranes may comprise at least one of a moveable cover, or a rotating enclosure surface. Each of the moveable pressure release valve members is configured to provide one-way flow out of battery pack **100**, while sealing against intrusion of water or other contaminants. Battery pack **100** includes multiple pressure release valves **108**, with two provided in front sidewall **102***a* and three located in rear sidewall **102***b*.

In contrast to plug valves **110** and pressure release valves **108**, deformable pressure release valves **112** may facilitate a relatively greater flow of at least one of pressurized or heated gas from within housing **124** (e.g., due to a sudden or extreme thermal event corresponding to at least one battery cell of at least one of battery modules **104**). Deformable pressure release valves **112** may include a deformable disc or other structure that is configured to mechanically deform (e.g., break, melt, or otherwise disintegrate such that deformable pressure release valves **112** form an opening in a sidewall of housing **124**) in response to a pressure or temperature above a predetermined amount (e.g., as may be characteristic of a thermal event of battery cells in battery pack **100** based on damage to or operation of battery modules **104**). Exemplary embodiment of pressure release valve **108** is illustrated in FIGS. **9**A and **9**B. Pressure is allowed to vent from battery pack **100** (e.g., as shown in FIG. **14**). Additionally, flow in the reverse direction, (e.g., into battery pack **100**) may be generally prevented by a perimeter seal arranged along at least one of an inner surface or an outer surface of sidewalls **102** (no shown). The perimeter seal may have a preload against a corresponding seal surface, which is generally defined by a flexibility of the perimeter seal (e.g., an ability to elastically deform and retain an original shape despite experiencing strain within an elastic range of the material comprising the perimeter seal) and an insertion load of pressure release valves **108** resulting in a seal between a perimeter of pressure release valves **108** and openings in sidewalls **102** configured to receive pressure release valves **108**.

Battery pack **100** may be configured to vent from within housing **124** (e.g., based on at least one of pressurized or heated gas generated within module bays **105** in response to operation of battery modules **104**) to the external environment surrounding housing **124** in response to different levels or thresholds of internal pressure or heat corresponding to the different fluid flow paths provided by valves **108**, **110**, and **112**, respectively. In one example, plug valves **110** may be configured to vent an internal pack pressure not exceeding a first pressure threshold (e.g., 5 kPa of pressure within the battery pack **100**) which may fall within a range of pressure created by thermal expansion of air contained within battery pack **100**. Plug valves **110** may also permit ingress of air in response to thermal contraction of air within battery pack **100**. Relatively higher levels of pressure (e.g., from the first pressure threshold to a second or higher pressure threshold), may be vented from battery pack **100** via pressure release valves **108** in response to venting of one or more battery modules **104**, higher levels of heat output of battery modules **104** based on the operating conditions of battery modules **104**, or other conditions creating additional pressure within battery pack **100**. In some embodiments, the pressure thresholds of the disclosure refer to pressure differentials between a space within the battery pack enclosing a battery module or a battery and a space external to the battery pack.

Accordingly, continuing with the example first pressure threshold introduced above, internal pressures of 5 kPa to 10 kPa within battery pack **100** may be vented by at least one of pressure release valves **108**. An internal pack pressure reaching a higher pressure threshold than can be serviced by plug valves **110** may be vented via pressure release valves **108**. Further, even higher levels of at least one of pressure or heat exceeding the second threshold level or pressure levels generally serviceable by pressure release valves **108** (e.g., above 10 kPa) may be vented from battery pack **100** by deformable pressure release valves **112**. In an example, a sudden onset of internal pressure may quickly reach the first pressure threshold pressure (e.g., 5 kPa) causing pressure release valves **108** to begin venting. If pressure continues to rise above levels serviceable by pressure release valves **108** (e.g., above 10 kPa or above 50 kPa), one or more of deformable pressure release valves **112** may mechanically deform, allowing further release of the internal pressure by creating an uncovered through opening in at least one of sidewalls **102** that enables egress from within module bays **105** to the environment surrounding housing **124**. Each of these outputs may propagate throughout battery pack **100**, as described further below. In some examples, a pressure buildup or flow exceeding a predetermined amount (e.g., above 10 kPa) may generate a warning for service of battery pack **100**. For example, control circuitry may be arranged within battery pack **100** which comprises sensors (e.g., a water sensor configured to detect standing water within the battery pack assembly, a temperature sensor, a voltage sensor, and a pressure sensor). These sensors may be configured to provide at least one of data or instructions to generate at least one warning to a vehicle operator, a vehicle controller, or any combination thereof.

Battery pack **100** is substantially sealed against egress of environmental fluids or materials, apart from the fluid flows permitted by the pressure release valves **108**, **110**, and **112** to address increases of at least one of pressure or temperature within sidewalls **102**. Sidewalls **102** and crossmembers **106** generally may create module bays **105** (e.g., individual enclosures within housing **124**) around at least one of battery modules **104** (e.g., in some embodiments two of battery modules **104** as shown in FIG. **1**A). While nine of battery modules **104** are illustrated being distributed amongst module bays **105**, any number of modules **104** or module bays **105** may be employed (e.g., contingent on a desired power output of battery pack **100** or at least one of a weight or space constraint associated with the environment surrounding housing **124**). Any or all of module bays **105** may contain any number of modules **104** that is convenient. Battery modules **104***a-i* may be comprised of a plurality of battery cells that are interconnected to generate an amount of electrical energy to be provided to a larger vehicle system (e.g., via at least one of a bus bar or at least one terminal connection). Battery modules **104***a-i* may be arranged vertically, horizontally, or may be stacked over each other depending on the available packing space of the structure for which the battery pack **100** is configured to provide electrical power. In some embodiments, battery modules **104** within one or more of module bays **105**, and in some cases each of module bays **105**, are positioned in a two-layer stack of battery cells. More specifically, as illustrated in FIG. **13**, a lower layer of battery modules may be positioned beneath an upper layer of battery modules. In some embodiments, a cooling apparatus or layer may be positioned between.

Module bays **105** of battery pack **100** may permit fluid communication via one or more of vent passages **118** to permit flow of at least one of pressurized or heated fluid (e.g., gas or liquid) to at least one of pressure release valves **108**, **110**, or **112** for venting. Vent passages **118** may be bidirectional (e.g., enabling ingress and egress of at least one of heated or pressurized fluid between two of module bays **105**) or multidirectional (e.g., enabling at least two of ingress, egress, or some other transitory propagation of at least one of heated or pressurized fluid between at least two of module bays **105** or between at least two of battery modules **104**, based on either a lateral or vertical arrangement of the at least two of battery modules **104**). Each of vent passages **118** are provided in laterally extending crossmembers **106** of battery pack **100**. Additionally, the forward-most module bay **105***a* may vent to the external environment surrounding housing **124** via at least one of pressure release valves **108**, and rearward most module bay **105***e* may vent to the external environment surrounding housing **124** via at least one embedded iteration of pressure release valves **108**.

Vent passages **118** are shown in FIG. **1**A as extending between different combinations of module bays **104** and may generally freely allow fluid flow. In some embodiments, vent passages **118** may also be configured to enable passage of heated or pressurized fluid between vertically arranged battery modules **104** (e.g., as shown in FIG. **13**). Accordingly, a thermal event or cell venting event in any of module bays **105** is communicated to an adjacent module bay of module bays **105**. Further, to the extent this may cause a buildup of pressure within module bays **105** collectively, resulting in pressure release valves **108** collectively venting at least one of pressurized or heated fluid (e.g., gas, vapor, or liquid) to the external environment surrounding housing **124**. Crossmembers **106** may provide structural support to at least one of battery pack **100** or a vehicle into which battery pack **100** is installed. Accordingly, vent passages **118** may generally facilitate a balance between the additional structural strength of battery pack **100** provided by crossmembers **106**, while also facilitating an adequate vent flow out of battery pack **100**. Vent passages **118** are positioned to facilitate vent flows along a floor structure (e.g., battery pack base **122** of FIG. **1**B). At least some battery cells comprising battery modules **104** may be positioned such that venting from the cells tends to flow toward the floor structure. As such, venting from these cells may be facilitated by vent passages **118**. At the same time, upper cells within the module may vent upwards within the module bays **104** toward a battery pack top or cover (e.g., battery pack cover **120**).

FIG. **1**B depicts an exploded view of battery pack **100** comprising battery pack cover **120** and battery pack base **122**. Battery pack base **122** provides a bottom support or surface of battery pack **100**, upon which at least one of battery modules **104**, various sensors, vehicle system interfaces and the like may be supported or arranged. Battery pack cover **120** is configured to enclose module bays **105** from above. Battery pack cover **120** may be positioned over mica sheets which would aid in managing heat and pressurized gas generated by the operation of battery cells (e.g., comprising battery modules **104** of FIG. **1**A) arranged in each of module bays **105**. Each of battery pack cover **120** and battery pack base **122** may be fixedly attached to at least a portion or at least one of any or all of sidewalls **102** and crossmembers **106**. In some embodiments, at least one pressure release valve may be arranged in a forward facing surface of sidewalls **102** while at least one pressure release valve may be arranged in at least one mounting location in a rear facing surface of sidewalls **102**. In some embodiments, battery pack **100** may comprise at least one additional sidewall **102**, that is not forward or rear facing, without at least one pressure release valve when at least one of a forward-facing wall of sidewalls **102** or a rear-facing wall of sidewalls **102** comprise enough deformable pressure release valves or plug valves to enable target pressurized or heated fluid egress rates.

Between each of module bays **105** is at least one of vent passages **118**, which may be positioned towards or at the bottom of each of a plurality of crossmembers **106**. The position of each of vent passages **118** in each respective crossmember of crossmembers **106** may depend on an optimal flow path for at least one of hot or pressurized fluid generated by the operation of battery modules **104**, as determined based on operating parameters of a particular embodiment of battery pack **100**. In some embodiments, at least one of vent passages **118** is unobstructed and at least one other of vent passages **118** is at least partially obstructed by various battery assembly components (e.g., coolant lines). In some embodiments, an inlet of at least one of vent passage **118**, enabling the egress of gas out of one of module bays **105**, is positioned higher than an outlet of the at least one vent passage, where a vent housing may be installed (e.g., a pressure release valve). In some embodiments, the inlet may be positioned lower than the outlet.

At least one of sidewalls **102** or crossmembers **106** may comprise an extruded material. At least one inlet or at least one outlet may comprise machined cutouts of the extruded material (e.g., to accommodate at least one of at least one vent passage **118** or at least one pressure release valve). Battery pack **100** may be formed by a method of manufacturing comprising extruding a rigid material to form at least one sidewall comprising an inner surface, an outer surface, and at least one guiding rib. The guiding rib may comprise a first end abutted to the inner surface, a second end abutted to the outer surface, and a profile structured to modify the trajectory of gas generated from inside the battery assembly. A center of the guiding rib is proximate to a vertical center of the sidewall such that the inlet and outlet maintain a target stiffness level of the inner surface and the outer surface. Once the at least one sidewall is formed, the at least one sidewall is arranged to form an enclosure at least partially. Within the enclosure, at least one battery module comprising a plurality of battery cells may be arranged. The sidewall may comprise a plurality of inlets and a corresponding plurality of outlets, wherein each of the plurality of inlets and corresponding outlets are adjacent to each other, to accommodate hot gas generated by a plurality of battery modules arranged in the enclosure. The described components and method of manufacturing may also be utilized to produce a battery pack, comprising at least one battery module comprising plurality of battery cells, and at least one sidewall arranged to encompass in part the battery module. The at least one sidewall may comprise an inner surface comprising an inlet opening configured to receive gas generated by the plurality of battery cells, an outer surface comprising an outlet opening configured to vent the gas, and a guiding rib that modifies a trajectory of gas between the inner surface and the outer surface.

FIG. **2**A depicts a cross sectional view of sidewall **200**A comprised of angled guiding ribs **202**A formed by material with a straight surface profile. FIG. **2**B depicts a cross sectional view of sidewall **200**B comprising curved guiding ribs **202**B, in accordance with some embodiments of the disclosure. Each of sidewalls **200**A and **200**B may comprise more or fewer than the components or features depicted in FIGS. **2**A and **2**B. Each of sidewalls **200**A and **200**B may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1** and **3-13**. Additionally, each of sidewalls **200**A and **200**B may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

Sidewall **200**A corresponds to any or all of sidewalls **102** of FIG. **1**, and may be arranged to form housing **124**. In some embodiments, sidewall **200**A may be positioned below a vehicle occupant zone or may be positioned to avoid venting at least one of pressurized or hot fluid from inside a battery pack onto components surrounding the battery pack (e.g., as shown in FIGS. **10**A, **10**B, and **10**C). Sidewall **200**A comprises inner surface **204**, outer surface **206**, and angled guiding ribs **202**A. Inner surface **204** is arranged and configured to receive at least one of hot or pressurized gas from inside a battery enclosure (e.g., module bays **105** of FIG. **1**) as generated by the operation of at least one battery module (e.g., battery modules **104** of FIG. **1**). Outer surface **206** comprises opening **208** configured to receive pressure release valve housing **210** affixed to outer surface **206** by interfacing with opening **208**. Pressure release valve housing **210** when interfacing with opening **208** comprises outlet **214**. Outlet **214** is configured to enable the egress of at least one of pressurized or hot gas after the pressurized or hot gas progresses through inlet **216** in response to a pressure differential between an environment in front of inlet **216** (e.g., corresponding to at least one of module bays **105** of FIG. **1**) and an environment outside of outlet **214** (e.g., a vehicle underbody). For example, the environment in front of inlet **216** may correspond to an atmospheric pressure that is below a pressure to the right of inner surface **204** of FIG. **2**. Between outer surface **206** and inner surface **204** are guiding ribs **202**A, which provide upper and lower boundaries for guiding at least one of hot or pressurized fluid (e.g., gas, vapor, or liquid) from inlet **216** through outlet **214**. In some embodiments, at least one of guiding ribs **202**A comprises a straight profile (e.g., as shown in FIG. **2**A) with first end **218** affixed to a surface that faces outlet **214**. First end **218** is arranged vertically higher than second end **220**, which is affixed a surface that faces inlet **216**. The steepness of at least one of guiding ribs **202**A and the profile prevents restriction of flow of at least one of hot or pressurized fluid by preventing turbulence in the flow of the hot or pressurized fluid and enables the hot or pressurized fluid to follow exit trajectory **222** instead of exit trajectory **224**. Exit trajectory **222** prevents the egress of the hot or pressurized fluid towards a portion of the environment external to outer surface **206** that contains components and the like that cannot withstand exposure to the hot or pressurized gas (e.g., as shown in FIGS. **10**A, **10**B, and **10**C). Exit trajectory **224** corresponds to an exit trajectory of the hot or pressurized fluid without the guiding ribs, which would result in the collection of the hot or pressurized fluid between inner surface **204** and outer surface **206**. Additionally, exist trajectory **224** corresponds to a fluid flow that may result in exposure of components and the like external to outer surface **206** that should not be exposed to the hot or pressurized fluid.

Sidewall **200**B comprises the components of sidewall **200**A without guiding ribs **202**A with a straight profile. Instead, sidewall **200**B comprises curved guiding ribs **202**B. The curved profile of guiding ribs **202**B begins with first end **218** affixed to a surface that faces outlet **214**. First end **218** is arranged vertically higher than second end **220**, which is affixed a surface that faces inlet **216**. The curved profile of guiding ribs **202**B prevents restriction of flow of at least one of hot or pressurized fluid by preventing turbulence in the flow of the hot gas while guiding the exit trajectory of the hot or pressurized fluid away from certain areas external to the battery assembly (e.g., surrounding components as shown in FIGS. **10**A, **10**B, and **10**C). The curve of the profile may be determined based on turbulence and restriction of flow target values for at least one of hot or pressurized fluid generated by the operation of battery modules (e.g., battery modules **104** of FIG. **1**) within an enclosure (e.g., module bays **105** of FIG. **1**). The change of direction of the fluid flow trajectory as defined by the curved profile of curved guiding ribs **202**B from inlet **216** to outlet **214** enables transition of vectors defining the fluid flow without creating turbulence in the fluid flow. In some embodiments, either or both of sidewalls **200**A or **200**B comprises extruded material, wherein inlet **216** and outlet **214** comprise machined cutouts of the extruded material. In some embodiments, sidewalls **200**A and **200**B have at least one guiding rib, wherein a center of the guiding rib is proximate to a vertical center of inner surface **204** and outer surface **206**, thereby maintaining a target stiffness level for either or both of sidewalls **200**A and **200**B. In some embodiments, at least one of sidewall **200**A or **200**B comprises a plurality of inlets **216** and outlets **214**, wherein each of the plurality of inlets **216** and outlets **214** are adjacent to each other in at least one of sidewall **200**A or **200**B (e.g., as shown in FIG. **3**).

FIG. **3** depicts sidewall assembly **300** comprising dual membrane pressure release valve **700** of FIGS. **7**A and **7**B, outlets **214** of FIG. **2**, and curved guiding ribs **202**B of FIG. **2**, in accordance with some embodiments of the disclosure. Sidewall assembly **300** may comprise more or fewer than the components or features depicted in FIG. **3**. Sidewall assembly **300** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1-2**B and **4-13**. Additionally, sidewall assembly **300** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

Sidewall assembly **300** comprises a plurality of outlets **214**, with dual membrane pressure release valve **700** of FIGS. **7**A and **7**B populating one of the plurality of outlets **214**. Exposed outlets **214** are shown for illustration and may be covered by additional dual membrane pressure release valves or other valves described herein. In some embodiments, all of the pressure release valves may comprise a single large membrane in place of the two separate membranes as shown in FIGS. **9**A and **9**B. Sidewall assembly **300** also comprises a plurality of curved guiding ribs **202**B of FIG. **2**. In some embodiments, sidewall assembly **300** comprises guiding ribs **202**A of FIG. **2**A. In some embodiments, a combination of guiding ribs **202**A and curved guiding ribs **202**B may be utilized. Sidewall assembly **300** may be arranged or positioned anywhere around housing **124** of FIG. **1** by being arranged where any or all of sidewalls **102** are depicted in FIG. **1**. Sidewall assembly **300** may at least partially form at least one of module bays **105** of FIG. **1**. Dual membrane pressure release valve **700** may be structured to increase a material stiffness of sidewall assembly **300**, considering the material stiffness of sidewall assembly **300** may be reduced by machining exposed outlets **214**. In some embodiments, at least one of plurality of dual membrane pressure release valves **700** may populated each of exposed outlets **214**. Each of the plurality of dual membrane pressure release valves **700** may correspond to a target stiffness value for the wall by being comprised of a housing that, when arranged within one of exposed outlets **214**, increases the material stiffness of sidewall assembly **300** to a target stiffness value (e.g., the yield or ultimate strength of sidewall assembly **300** increased when at least one of exposed outlets **214** is populated by one of dual membrane pressure release valves **700**). For example, sidewall **300** with exposed outlets **214** may have a reduction of at least 15% in the material stiffness of material comprising sidewall **300**. When exposed outlets **214** are populated by dual membrane pressure release valves **700**, the material stiffness of sidewall **300** may only be reduced by 5%, may return to an initial material stiffness, or may exceed the material stiffness of sidewall **300** before exposed outlets **214** are machined.

FIG. **4** depicts view **400** with sidewall assembly **300** of FIG. **3**, comprising at least one pressure release valve configured to enable rapid egress of at least one of pressurized or hot fluid generated within a battery assembly enclosed at least partially by sidewall assembly **300**, arranged to vent the pressurized or hot fluid towards target egress area **402** and away from protected area **404**, in accordance with some embodiments of the disclosure. View **400** may comprise more or fewer than the components or features depicted in FIG. **4**. View **400** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1-3** and **5-13**. Additionally, the components depicted in or described in reference to view **400** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

As shown in view **400**, sidewall assembly **300** comprises a plurality of outlets **214** of FIG. **2**. Exit trajectory **222** corresponds to the exit trajectory of at least one of pressurized or hot fluid, as shown in and described in reference to FIG. **2**. Exit trajectory **222** is shown as being directed towards target expulsion area **402**. Target expulsion area **402** corresponds to an area which outlets **214** guides the at least one of pressurized or hot fluid towards without exposing components or areas around sidewall assembly **300** (e.g., areas around housing **124** of FIG. **1**) that are not configured or structured to withstand the hot or pressurized fluid. For example, FIGS. **10**A, **10**B, and **10**C each provide structural areas and components that may be arranged out of target expulsion area **402** to avoid exposure to the expelled pressurized or hot fluid. Sidewall assembly **300** is positioned or arranged below protected area **404**.

Protected area **404** may correspond to a vehicle body, a vehicle frame, an occupant area, a cargo area, or any suitable form of enclosure, structural component, or assembly that is configured to interface or couple with a battery pack or battery assembly comprised of sidewall assembly **300**. View **400** includes vehicle assembly **408**, which is inclusive of all the elements of FIG. **4**. Protected area **404** is not arranged or configured to receive any portion of exit trajectory **222** as protected area **404** is arranged above top lateral edge **406**. Exit trajectory **222** may be enabled by at least one of a pressure release valve (e.g., a single or dual membrane valve as shown in FIGS. **7**A-**9**B) or at least one guiding rib connecting surfaces forming sidewall assembly **300** (e.g., as shown in FIGS. **2**A-**3**). Exit trajectory **222** enables the expulsion of at least one of pressurized or hot fluid downward award from top lateral edge **406**. Additional single or dual membrane pressure release valves may be positioned in sidewall assembly **300** or a battery pack assembly comprised of housing **124** of FIG. **1** to optimize the amount of hot or pressurized fluid expelled during any given event which causes a buildup of pressurized gas within the battery pack assembly.

FIG. **5** depicts a cross sectional view of sidewall assembly **500**, in accordance with some embodiments of the disclosure. Sidewall assembly **500** may comprise more or fewer than the components or features depicted in FIG. **5**. Sidewall assembly **500** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1-4** and **6**A-**13**. Additionally, sidewall assembly **500** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

Sidewall assembly **500** comprises first guiding rib **506**A, wherein first end **508**A of first guiding rib **506**A is positioned towards or substantially at top edge **510**A of the opening of inlet **216**. Second end **508**B of first guiding rib **506**A is positioned towards or substantially at top edge **510**B of the opening of outlet **214**. Sidewall assembly **500** also comprises second guiding rib **506**B, wherein first end **512**A of second guiding rib **506**B is positioned towards or substantially at bottom edge **514**A of the opening of inlet **216**. Second end **512**B of second guiding rib **506**B is positioned towards or substantially at bottom edge **514**B of the opening of outlet **214**. Sidewall assembly further comprises third guiding rib **506**C. Third guiding rib **506**C is positioned or arranged vertically between first guiding rib **506**A and second guiding rib **506**B. Third guiding rib **506**C forms first channel **516**A with first guiding rib **506**A and second channel **516**B with second guiding rib **506**B, wherein each of first channel **516**A and second channel **516**B provide an open egress path for at least one of hot or pressurized fluid with a trajectory flowing towards an outer surface of inlet **216** and towards an inner surface of outlet **214**.

Third guiding rib **506**C is comprised of first recess **518**A arranged towards the opening of inlet **216**, wherein first recess **518**A is configured to reduce turbulence corresponding to the trajectory of gas. First recess **518**A may be at least one of configured to enable gas flow at target turbulence levels or incorporated for ease of manufacturing or assembly. By reducing inlet area with the recess, inlet **216** is configured to reduce a likelihood of restricted flow and improve efficiency of flow when a rapid egress of at least one of hot or pressurized fluid is required to maintain function of a battery assembly. The recess may also reduce the surface across which the flow of hot gas travels and also reduces the likelihood of turbulence in order to improve the efficiency of the egress of the at least one of hot or pressurized fluid. Third guiding rib **506**C also is comprised of second recess **518**B arranged towards the opening of outlet **214**, wherein second recess **518**B is configured to accommodate a rear portion of a valve affixed to the outer wall (e.g., membrane **504** in vent housing **210**). Second recess **518**B is arranged such that neither a surface nor edge of material forming second recess **518**B contacts a rear portion of the pressure release valve comprising membrane **504**. In some embodiments, an accommodating recess is not formed in third guiding rib **506**C towards inlet **216**.

Affixed an outer surface of sidewall assembly **500**, corresponding to an outer surface of outlet **214**, is vent housing **210**. Vent housing **210** may comprise a membrane pressure release valve configured to deform when exposed to hot gas corresponding to a thermal runaway condition such that membrane **504** of the pressure release valve comprising vent housing **210** and membrane **504** are mechanically displaced (e.g., mechanically releasing a seal formed between the membrane and the housing), creating a large opening at outlet **214** for the rapid egress of at least one of hot or pressurized gas. Embedded in a groove of vent housing **210** is vent gasket **502**. Vent gasket **502** is configured to mechanically seal against a surface of sidewall assembly **500** such that at least one of hot or pressurized gas is directed to outlet opening **514**B. Additionally, vent gasket **502** is configured to prevent moisture from entering sidewall assembly **500** through vent housing **210**.

FIG. **6**A depicts first sidewall **600**A and FIG. **6**B depicts second sidewall **600**B, in accordance with some embodiments of the disclosure. First sidewall **600**A and second sidewall **600**B may comprise more or fewer than the components or features depicted in FIGS. **6**A and **6**B. First sidewall **600**A and second sidewall **600**B may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1-5** and **7A-13**. Additionally, at least one of first sidewall **600**A or second sidewall **600**B may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

First sidewall **600**A is comprised of angled vent housing **602** that is embedded (e.g., affixed) in an outer surface of first sidewall **600**A. It will be understood that vent housing **602** is angled with respect to the outer surface (e.g., rotated about an axis normal to the outer surface), such that one side of vent housing **602** is lower than the other side of vent housing **602**. First sidewall **600**A comprises a pair of surfaces, the outer surface configured to receive angled vent housing **602** and an inner surface comprised of inlet **214**. Between the outer surface and the inner surface is internal cavity **604** for collecting moisture such that it does not enter an enclosure partially formed by first sidewall **6001** (e.g., module bays **105** of FIG. **1**). Collected moisture **606** corresponds to fluids that was able to enter between the inner and outer surfaces of first sidewall **600**A. Preferably, collected moisture **606** should have a volume substantially equivalent to zero (e.g., no moisture enters first sidewall **600**A or second sidewall **600**B). However, it may be understood that during operation of a battery pack comprised of at least one of first sidewall **600**A or second sidewall **600**B various sealing interfaces may have imperfect sealing function and moisture may collect at various interfaces throughout the battery pack. As a result, the battery pack comprised of at least one of first sidewall **600**A or second sidewall **600**B may be configured to remain operable despite moisture collecting within the assembly. For example, a maximum level of allowable moisture (e.g., a maximum fluid ingress amount) in a battery enclosure may be 1.5 mm. Internal cavity **604** may be structured such that internal cavity **604** can store enough moisture collected from the environment around the battery pack or during operation of the battery pack that 1.5 mm of moisture is prevented from collecting in the enclosure at least partially formed by at least one of first sidewall **600**A and second sidewall **600**B (e.g., the height of the inner surface and outer surface forming the internal enclosure is configured such that a volume of water corresponding to at least 1.5 mm throughout the enclosure can be stored in the internal cavity). Second sidewall **600**B includes openings in an outer surface for the installation horizontally aligned vent housing **608**. Second sidewall **600**B also comprises inlets **214**, each corresponding to at least one of vent housings **608**. First sidewall **600**A may be arranged such that an outlet of angled vent housing **602**A faces a forward portion of a vehicle assembly comprised of a battery pack comprised of first sidewall **600**A. Second sidewall **600**B may be arranged such that the outlets of horizontally aligned vent housings **608** face a rear portion of a vehicle assembly comprised of a battery pack comprised of second sidewall **600**B. It will be understood that there may be more or fewer of at least one of venting openings or venting housings, depending on the desired egress rates of at least one of pressurized or heated fluid generated through the operation of a plurality of battery cells operating proximate to the depicted sidewalls (e.g., within module bays **105** of FIG. **1**). The venting housings may comprise dual or single membrane pressure release valves, depending on the desired preload-sealing criteria of a particular battery assembly.

FIG. **7**A depicts outer surface **702**A of dual membrane pressure release valve **700** and FIG. **7**B depicts inner surface **702**B of dual membrane pressure release valve **700**, in accordance with some embodiments of the disclosure. Dual membrane pressure release valve **700** may comprise more or fewer than the components or features depicted in FIGS. **7**A and **7**B. Dual membrane pressure release valve **700** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1-6** and **8**A-**13**. Additionally, dual membrane pressure release valve **700** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

Dual membrane pressure release valve **700** comprises housing **704** with rigid mounts **706**. In some embodiments, rigid mounts **706** comprise features that extend from housing **704** and may interface with or at least partially comprise rigid mounts **706** such that dual membrane pressure release valve **700** may receive or be configured to interface with external deflectors. For example, each of rigid mounts **706** may corresponds to or may be comprised of extensions configured to form a snug fit interface (e.g., based on a compression fit specification that requires a press in load to be 50% or less of a pull out load once the interface is established by pressing in each of rigid mounts **706** into an opening in the sidewall) with an opening in the sidewall corresponding to the outline of housing **704** that is inclusive of the extensions or rigid mounts. Each of rigid mounts **706** incorporate at least one of rigid inserts **708** to reduce a transfer of torque or load from a fastener configured to interface with each of rigid mounts **706** to other features of housing **704** (e.g., to prevent mechanical deformation of housing **704** which may compromise sealing around dual membrane pressure release valve **700** by resisting a compression force generated by applying an installation torque to the fastener from transferring to the rigid mount). In some embodiments, rigid inserts **708** may be comprised of a metal material while housing **704** is comprised of a plastic material with a lower material strength than the metal material used for rigid inserts **708**. Extending features **710** encompass each of membranes **712** to increase stiffness of housing **704** in order to prevent mechanical deformation (e.g., elastic or plastic) of housing **704**.

Each of membranes **712** comprise an area corresponding to a target preload required to translate membranes **712** to enable the egress of at least one of hot or pressurized fluid from behind each of membranes **712** (e.g., as generated by the operation of battery modules in module bays **105** of FIG. **1**). Membranes **712** are configured to create a seal against respective openings in housing **704**. The openings correspond to an area formed at least partially by deformable struts **722**. The target preload of each of membranes **712** may also correspond to target sealing levels to prevent the ingress of moisture behind the membranes (e.g., reducing an amount of fluid collected in FIG. **6**). For example, the total area of membranes **712** may correspond to a maximum flow rate of the egress of at least one of pressured or heated fluid to prevent a buildup of pressure or heat in a battery pack comprised of sidewalls with at least one of dual membrane pressure release valve **700** embedded in an outer surface of at least one of the sidewalls. In order to balance the egress of hot gas against the need to prevent a threshold amount of moisture from entering the battery pack, the total area of membranes **712** is divided between two separate membrane structures such that each of membranes **712** corresponds to a high enough sealing preload to prevent the ingress of moisture without exceed a sealing preload that would prevent rapid egress of at least one of pressurized or heated fluid. The sealing preload corresponds to a force required to translate at least one of membranes **712** away from housing **704** to create an opening outer surface **702**A. Forces acting on a battery pack comprised of dual membrane pressure release valve **700** may affect the integrity of a seal formed by each of membranes **712**. For example, various forms of excitation to the surfaces of the battery pack caused by pressure differentials, vibrating mediums, or rapid changes in direction may comprise a seal by disrupting the contact surfaces along each of membranes **712**. As a result, the battery pack may be configured to remain operational with some fluid ingress that is below a threshold fluid ingress that may affect operation of the battery pack.

Inner surface **702**B is arranged to face opposite of outer surface **702**A. As shown, rigid mounts **706** are arranged around the periphery of housing **704**. Each of rigid mounts **706** are comprised of reinforcement vanes **714** which are structure to increase the material stiffness of each of rigid mounts **706**, thereby reducing a risk of mechanical deformation or deflection during installation or articulation of membranes **712**. Reinforcement valves **714** are arranged radially around openings in each of rigid mounts **708** to prevent deflection or deformation of rigid mounts **708**. Rigid mounts **708** may comprise a first rigid mount arranged to protrude from a top side of housing **702** and further arranged laterally between membranes **712**, a second rigid mount arranged to protrude from a bottom of housing **704** and further arranged to protrude from a first side of housing **704** (e.g., from a first bottom corner of housing **704**), and a third rigid mount arranged to protrude from a bottom of housing **704** and further arranged to protrude from a second side of housing **704** (e.g., from a second bottom corner of housing **704**).

Housing **704** is comprised of gasket groove **716** in which gasket **718** is arranged to create a seal between housing **704** and a surface of a battery pack sidewall in which housing **704** is installed or embedded. Gasket **718** is comprised of gasket protrusions **720**, which are structured to secure gasket **718** in gasket groove **716** by increasing friction between gasket **718** and gasket groove **716**. Gasket **718** encircles membranes **712**, which are supported by deformable struts **722**. Gasket **718** extends partially out gasket groove **716** such that when rigid mounts **708** are secured to a mounting sidewall, gasket **718** is partially compressed against the mounting sidewall and acts as a seal to prevent hot or pressurize fluid from passing between the mounting sidewall and housing **704**. Gasket groove **716** may further comprise a pair of recesses between membranes **712** configured to accommodate gasket protrusions **720**. Gasket **718** may comprise a plurality of additional protrusions structured to match a profile of gasket groove **716** in housing **704** such that gasket **718** remains in a secured position and maintains a continuous seal encircling both of membranes **712**. Deformable struts **722** are configured to mechanically deform and release membranes **712** (e.g., the membranes are displaced or fall out of their respective housings) when exposed to thermal runaway conditions, thereby creating openings in housing **704** where membranes **712** are depicted in FIGS. **7**A and **7**B. The housing may comprise three rigid mounts as shown.

FIG. **8**A depicts outer surface **802**A of pressure release valve **800** and FIG. **8**B depicts inner surface **802**B of pressure release valve **800**, in accordance with some embodiments of the disclosure. Pressure release valve **800** may comprise more or fewer than the components or features depicted in FIGS. **8**A and **8**B. Pressure release valve **800** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1-7**B and **9**A-**13**. Additionally, pressure release valve **800** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

Pressure release valve **800** comprises extended membrane **804**, which comprises an area which avoids a need to include a second membrane, in accordance with some embodiments of the present disclosure. Pressure release valve **800** is comprised of deformable struts **806** (e.g., as shown in FIG. **8**B), rigid mounts **706** (e.g., configured to receive rigid inserts and comprised of reinforcement vanes as shown in FIG. **7**), and gasket groove **716** configured to receive a sealing gasket. Additionally, housing **708** comprises at least one of stiffening extension **810** such that housing **708** corresponds to a material stiffness that does not compromise or create a weak point in a sidewall in which pressure release valve **800** may be installed.

The pressure release valves shown in FIGS. **7**A-**8**B may be incorporated into a battery pack. The battery pack may comprise elements and the assemblies depicted in any or all of FIGS. **1**-**13**. The battery pack may comprise at least one sidewall, wherein the sidewall is arranged to form an enclosure at least partially around a battery module, and wherein the at least one sidewall comprises at least one outlet. The battery module may be positioned laterally proximate to the at least one sidewall, wherein the battery module comprises a plurality of battery cells. The pressure release valves of FIGS. **7**A-**8**B may comprise at least one venting structure (e.g., membranes **712** or **804**), wherein the at least one venting structure is fixedly attached to the at least one outlet. The at least one venting structure may comprise a housing configured to be secured to the at least one sidewall, a first membrane positioned towards a first side of the housing, a second membrane positioned adjacent to the first membrane and positioned towards a second side of the housing, and a gasket structured to form a continuous seal between the housing and the sidewall.

FIG. **9**A depicts battery pack **900** comprised of sidewall **902** with external surface **904**, in accordance with some embodiments of the disclosure. FIG. **9**B depicts inner surface **906** of sidewall **902**, in accordance with some embodiments of the disclosure. Battery pack **900** may comprise more or fewer than the components or features depicted in FIGS. **9**A and **9**B. Battery pack **900** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1**A-**8**B and **10**A-**13**. Additionally, battery pack **900** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

Sidewall **902** is shown with dual membrane pressure release valves embedded in external surface **904**, in accordance with some embodiments of the present disclosure. Sidewall **902** comprises top lateral edge **912**. Dual membrane valves **700** are arranged in positions configured optimize the egress of at least one of pressurized or heated fluid from inside battery pack **900**. Each of dual membrane pressure release valves **700** is depicted as being rotated or moved towards the center of external surface **904** (e.g., at the depicted angle α or along the trajectories illustrated by the arrows), depending on where vehicle components around battery pack **900** are arranged or where the optimal egress position for the fluid is, considering the arrangement of battery cells or battery modules and other components inside battery pack **900** and surrounding structures. In some embodiments, single membrane pressure release valve **908** is embedded in angled sidewall **910**, which is fixedly attached to external surface **904** of sidewall **902** to at least partially form at least one enclosure for battery cells (e.g., module bays **105** of FIG. **1**). An additional single membrane pressure release valve (not shown) may also be embedded in an angled sidewall fixedly attached on an opposite side of external surface **904** of sidewall **902** to balance the release of at least one of pressurize or heated fluid across sidewall **902**. The cumulative area of single membrane pressure release valve **908** and dual membrane pressure release valves **700** may equal or correspond to the cumulative area of pressure release valves installed in a rear-facing sidewall of the battery assembly. For example, there may be up to a 5% difference in area between the cumulative area of single membrane pressure release valve **908** and the cumulative area of dual membrane pressure release valves **700**. In some embodiments, the vehicle assembly may result in a functional need for either sidewall **902** or a rear-facing sidewall arranged to face opposite of external surface **904** to serve as a primary egress opening for at least one of pressurized or heated fluid, depending on which sidewalls are arranged to have outlets which are directed towards target expulsion area **402** of FIG. **4**. Dual membrane pressure release valves **700** are optimally positioned on external surface **904** such that the stiffness of sidewall **902** maintains a target stiffness (e.g., near the vertical center of sidewall **902**).

FIGS. **10**A, **10**B, and **10**C each depict battery pack **100** with dual membrane pressure release valves **700** embedded in a sidewall, in accordance with some embodiments of the disclosure. Battery pack **100** may comprise more or fewer than the components or features depicted in FIGS. **10**A, **10**B, and **10**C. Battery pack **100** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1**A-**9**B and **11**A-**13**. Additionally, battery pack **100** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

As shown in FIG. **10**A, dual membrane pressure release valves **700** are positioned such that when the pressure release valves enable the egress of at least one of pressurized or heated fluid from within battery pack **100**, the trajectory of the expelled fluid does not get influenced or modified by the depicted features surrounding battery pack **100**. Additionally, the positioning of dual membrane pressure release valves **700** is influenced by impact area **1002**. Impact area **1002** corresponds to a portion of the external surface of battery pack **100** that is structured to receive impacts of surrounding components in response to a vehicle impact event or other related event (e.g., an event where a rapid deceleration occurs resulting in the deformation of vehicle components surrounding battery pack **100**). Dual membrane pressure release valves **700** are arranged such that the target stiffness of the sidewall of battery pack **100** is not compromised by the presence of the pressure release valve and neither of dual membrane pressure release valves **700** is subjected to a direct component impact in the aforementioned event or events. As shown in FIGS. **10**A, **10**B and **10**C, dual membrane pressure release valves **700** are mounted with their inboard sides rotated upwards (e.g., at least one rigid mount is arranged closer to a top lateral edge of a sidewall of battery pack **100** than at least one other rigid mount), which raises the lowest point (e.g., the lowest rigid mount) of the pressure release valves. In some embodiments, the lowest point of the pressure release valves is raised to provide clearance between an edge of either of dual membrane pressure release valves **700** or the outermost edge of impact area **1002**.

In FIG. **10**B, vehicle component profile **1004**A is positioned relatively close to a periphery of sidewalls forming battery pack **100**. There may be additional clearance between the forward-facing portion of battery pack **100** and vehicle component profile **1004**A to ensure impact area **1002** is not subjected to unexpected impacts from surrounding components. Accordingly, the location of dual membrane pressure release valves **700** on the forward-facing portion of the sidewall enables expelled fluid to substantially avoid vehicle component profile **1004**A. In FIG. **10**C, vehicle component profile **1004**B extends forwards from underneath a front sidewall of battery pack **100** (e.g., a z-bracket that provides support for the front of the battery assembly). Vehicle component profile **1004**B is centered with respect to battery pack **100** and the lateral locations of dual membrane pressure release valves **700** are selected such that expelled hot gas substantially avoids the component occupying a space substantially similar to vehicle component profile **1004**B.

FIGS. **11**A and **11**B depict a pair of views of battery pack **100** of FIG. **1**, in accordance with some embodiments of the disclosure. Battery pack **100** may comprise more or fewer than the components or features depicted in FIGS. **11**A and **11**B. Battery pack **100** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1**A-**10**C, **12**, and **13**. Additionally, battery pack **100** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

Multidirectional vent passages **118** remain generally unobstructed by components of battery pack **100**, and as a result freely permit fluid exchanges between adjacent bays **105** of the battery pack **100** at least along floor structure **114**. Additional nominal fluid exchange between bays **105** may nevertheless occur via other pathways within battery pack **100**. For example, coolant passages extend in a longitudinal direction through battery pack **100**, extending through corresponding crossmember apertures **134** of one or more of crossmember(s) **106**. Crossmember apertures **134** may generally fit closely around coolant passages **132**, permitting a nominal amount of fluid exchange between adjacent bays **105**. Further, a tunnel structure of battery pack **100** may extend longitudinally along a lateral center of battery pack **100**, resulting in additional nominal pathways for fluid exchange between bays **105** (e.g., over a top part of the crossmembers **106** along cover **116** which is not shown in FIGS. **11**A and **11**B). In contrast to these pathways for nominal fluid exchange between bays **105**, multidirectional vent passages **118** provide a relatively unobstructed and direct path for fluid exchange between bays **105**. Additionally, to the extent modules **104** may have batteries/cells venting downward toward floor structure **114**, multidirectional vent passages **118** provide a direct path between bays **105** coinciding with the direction of venting of these module(s) **104**.

The bidirectional vent passages may have any shape or configuration that is convenient. Multidirectional vent passages **118** may be positioned at a lower portion of each of bays **105**, adjacent floor structure **114** of battery pack **100**. Multidirectional vent passage **118** may be relatively wider along floor structure **114** (e.g., having a width at least twice as great as a height thereof). One or more of the bays **105**, or even each of bays **105** of battery pack **100**, may include two or more adjacent battery modules **104**, with crossmembers **106** on either side of modules **104** defining respective multidirectional vent passages **118**. Any number of crossmembers **106** may be employed, defining any number of bays **105**, as noted above. Multidirectional vent passages **118** of crossmembers **106** are each aligned longitudinally, thereby providing a straight and unobstructed flow path between each of bays **105**. Additionally, the battery modules **104** within one or more of the bays, or within each bay **105** of battery pack **100**, may be installed within battery pack **100** such that modules **104** are raised above floor structure **114**, thereby allowing fluid flow within each bay **105** along floor structure **114** beneath battery modules **104**.

The restricted flow paths to a number of pressure release valves **108** at the front and rear locations may facilitate venting from battery pack **100** at desired locations (e.g., with respect to a vehicle in which battery pack **100** is installed). For example, the locations of pressure release valves **108** may correspond to locations of the vehicle where a flow of heat or pressure may be relatively unnoticed by passengers, or otherwise may not impact vehicle systems. In some embodiments, at least one pressure release valve is embedded in a first sidewall positioned in a first portion of a vehicle comprised of battery pack **100**, wherein the first portion may be a front portion. In some embodiments, at least one pressure release valve is embedded in a second sidewall positioned in a second portion of the vehicle, wherein the second portion may be a rear portion. Either the first or the second portion may be either a front or a rear portion. By way of example, rearward pressure release valves **108** may vent to an area outer to the vehicle and beneath the vehicle, near a front or rear cargo area, cargo box, or the like. Accordingly, passengers exiting the vehicle or standing near the vehicle may be unlikely to notice the venting. Focusing the venting locations to the front and rear of battery pack **100** may also facilitate redirection of a vent flow or heat to areas of the vehicle away from temperature-sensitive areas or materials. In some examples, the vehicle may have deflectors or other devices for redirecting vent flow form the battery pack **100**.

The restricted flow paths for venting battery pack **100** may also facilitate consistent venting of valves **108** at a desired pressure level. More particularly, relatively larger vents may typically be more difficult to control with respect to providing consistent venting at a desired pressure level. For example, for larger pressure release valve structures it may be difficult to achieve a sufficient preload that will permit venting at the appropriate internal pressure, prevent venting at lower pressures where venting is not necessary, and also sufficiently seal against external contaminants such as water. Accordingly, in the example illustrations shown, there are multiple pressure release valves **108** along the front and rear venting locations. As a result, the pressure release valves **108**, and in particular moveable umbrella structures **109** are each relatively small and thus may more consistently provide venting at a desired pressure. As noted above, pressure release valves **108** may be configured to vent in response to an internal pressure of 5 kPa.

FIG. **11**B depicts battery pack **100** comprising module bays **105** formed by crossmembers **106**, each comprising gas egress openings in accordance with some embodiments of the present disclosure. Also depicted is rear-facing sidewall **102***b* comprising a plurality of dual membrane pressure release valves **108**. Dual membrane pressure release valves **108** are embedded in the rear-facing sidewall **102***b* and may be laterally aligned. In some embodiments, three pressure release valves comprise the plurality of pressure release valves. Each of the three pressure release valves are distanced by an equal amount from each other and arranged to enable the egress of at least one of pressurize or heated fluid from inside module bays **105** such that the egress results in fluid trajectories that guide the fluid around critical vehicle components. In some embodiments rear facing sidewall **102***b* may comprise one or more guiding ribs, as described above, to modify the exit trajectory of gas that enters sidewalls **102** and passes through dual membrane pressure release valves **108**. Rear facing dual membrane pressure release valves **108** may be configured to complement at least one pressure release valve embedded in forward-facing sidewall **102***a*, which is arranged at an opposite end of battery pack **100**. The cumulative area of the pressure release valves may correspond with target gas egress rates and a target gas egress direction (e.g., towards the forward-facing sidewall or towards the rear-facing sidewall to avoid venting gas in the direction of critical vehicle components or occupant arears).

FIG. **12** depicts battery pack **1200**, in accordance with some embodiments of this disclosure. Battery pack **1200** may comprise more or fewer than the components or features depicted in FIG. **12**. Battery pack **1200** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1**A-**11**B and **13**. Additionally, battery pack **1200** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

Battery pack **1200** is comprised of sidewall **102**D of FIG. **1**, and angled sidewall **1202** (e.g., angled at angle α relative to a plane defined by forward-facing sidewall **102**A that enables the egress of gas away from critical vehicle components), which at least partially form a plurality of battery cell enclosures (e.g., module bays **105** of FIG. **1**). Single membrane pressure release valves **1204** may be installed in each of single membrane pressure release valve outlets **1206**. In some embodiments, there is at least one single membrane pressure release valve per sidewall per battery cell enclosure. In some embodiments, valve outlets **1206** are not present (e.g., having less than four present or zero) along the sides of battery pack **1200** as the venting enabled by forward-facing sidewall pressure release valves **1208** and rear-facing sidewall pressure release valves **1210** enable adequate venting of battery pack **1200**. In some embodiments, dual membrane pressure release valves may be utilized, depending on target gas egress rates balances against moisture sealing requirements and sidewall stiffness requirements. Also depicted are forward-facing sidewall pressure release valves **1208** and rear-facing sidewall pressure release valves **1210**. Each of the forward and rear-facing pressure release valves are configured to complement the pressure release valves of the opposite facing sidewall. The cumulative area of the pressure release valves may correspond with target gas egress rates and a target gas egress direction (e.g., towards the forward-facing sidewall or towards the rear-facing sidewall to avoid venting gas in the direction of critical vehicle components or occupant arears).

FIG. **13** shows battery pack **1300**, in accordance with some embodiments of the disclosure. Battery pack **1300** may comprise more or fewer than the components or features depicted in FIG. **13**. Battery pack **1300** may be incorporated into or may incorporate any or all of the components or features depicted in or described in reference to FIGS. **1**A-**12**. Additionally, battery pack **1300** may be assembled, developed, or manufactured based at least in part on any of the steps depicted in or described in reference to FIG. **14**.

FIG. **13** shows a cross section of battery pack **1300** which enables the merging of fluid trajectories **1310** characterized by the arrows between battery modules **104**, based on the battery assembly configuration which as shown may include vertically stacked battery modules, to define target outlet location **1302** in a sidewall at least partially defining a battery enclosure (e.g., module bays **105**). Target outlet location **1302** is configured to be at a heigh in sidewall **102** corresponding to a maximum flow rate. For example, target outlet location **1302** may be at a height on sidewall **102** that is between 20% and 80% of the overall height of sidewall **102**. Battery pack **1300** comprises battery pack cover **1304** and battery pack base **1306** fixedly attached to at least one of enclosure crossmember **106** and at least one of sidewall **102**. Crossmember **106** may comprise gas egress opening **1308** corresponding to a fluid flow trajectory that contributes to fluid trajectories **1310**. Sidewall **102** may comprise a venting structure at an optimal height corresponding to target outlet location **1302**. The venting structure may comprise a plurality of pressure release valves (e.g., two pressure release valves angled on a forward-facing end sidewall, or three pressure release valves laterally aligned on a rear-facing end sidewall). The cumulative area of the pressure release valves may correspond with target gas egress rates and a target gas egress direction (e.g., towards the forward-facing sidewall or towards the rear-facing sidewall to avoid venting gas in the direction of critical vehicle components or occupant arears). The venting structure may be positioned in sidewall **102** at an optimal height based on fluid trajectories **1310**. Fluid trajectories **1310** may correspond at least one of hot or pressurized fluid generated by the operation of the depicted battery modules **104** separated by crossmember **106**. Battery pack **1300** may further comprise at least one of mica barrier **1312** positioned above each of a plurality of enclosures formed at least partially by sidewall **102** and crossmember **106** between battery pack cover **1304** and battery pack base **1306**. Battery pack cover **1304** and battery pack base **1306** may be fixedly attached to a base of sidewall **102** (e.g., a forward-facing sidewall and or a rear-facing sidewall). Battery modules **104** arranged on top of the other modules may be positioned such that pressurized or heated fluid vents upward towards battery pack cover **1304** while the lower layer of modules **104** may be positioned to vent downwardly toward battery pack base **1306**.

FIG. **14** is a flowchart of method **1400** for at least one of manufacturing or venting a battery assembly (e.g., battery pack **100** of FIG. **1**), in accordance with some embodiments of the disclosure. Method **1400** may comprise more or fewer than the steps depicted in FIG. **14**. Method **1400** may be incorporated into the development, assembly, manufacturing, or venting of any or all of the components or features depicted in or described in reference to FIGS. **1**A-**13**. Process blocks depicted with dashed borders comprise steps that are optional in developing, assembling, manufacturing, or venting battery pack **100** of FIG. **1**.

At **1402**, at least one sidewall comprising an inner wall, an outer wall, and at least one guiding rib is provided, wherein the at least one guiding rib comprises a first end that abuts the inner wall, a second end that abuts the outer wall, a profile structured to modify the trajectory of gas generated from inside the battery assembly. In some embodiments at **1402**A, providing at least one sidewall comprises extruding a rigid material to form the at least one sidewall. In some embodiments, method **1400** further comprises machining, at **1402**B, an inlet opening in the inner wall to expose a first side of the guiding rib. In some embodiments, at **1402**C, an outlet opening is machined in the outer wall to expose a second side of the guiding rib, wherein machining the second opening comprises machining an accommodating recess for a vent structure. At **1404**, the at least one sidewall is arranged to form an enclosure at least partially. At **1406**, a battery module is arranged inside the enclosure, wherein the battery module comprises a plurality of battery cells. In some embodiments, at **1408**, the vent structure is mounted to the outer wall such that the vent structure is aligned with the outlet opening.

In some embodiments, the battery assemblies described herein comprise sidewalls manufactured by a particular method. For example, a rigid material (e.g., aluminum), may be extruded into a shape that comprises at least one guiding rib. The guiding rib may comprises a profile corresponding to a target gas trajectory and wherein the at least one guiding rib interfaces with a first side of the extruded rigid material at a first location on the first side which is higher than where the guiding rib interfaces with a second side of the extruded rigid material. A first opening may be machined (e.g., milled or drilled) in the first side of the extruded rigid material to expose a first side of the guiding rib, wherein the first opening is positioned higher than a second opening in a second side. A second opening may also be machined (e.g., milled or drilled) in the second side of the extruded rigid material to expose a second side of the guiding rib, wherein machining the second opening comprises machining an accommodating recess for a vent structure. A vent structure may then be mounted to the second side of the sidewall. such that the vent structure is aligned with the second opening;

In some embodiments, the battery packs described herein may have venting structures or pressure release valves positioned in the sidewalls of the battery packs by a particular method of manufacture. A frame assembly may be provided for the battery pack. The frame assembly may comprise a plurality of sidewalls, wherein the plurality of sidewalls further comprises at least a forward-facing sidewall and a rear-facing sidewall. At least one of the battery pack or the frame assembly may comprise any number of additional components described herein or shown in FIGS. **1A-13**. A first plurality of pressure release valves may be mounted on the forward-facing sidewall and a second plurality of pressure release valves may be mounted to the rear-facing sidewall. The pressure release valves may be two membrane pressure release valves, single membrane pressure release valves, and any combination thereof. The cumulative area of the pressure release valves may correspond with target gas egress rates and a target gas egress direction (e.g., towards the forward-facing sidewall or towards the rear-facing sidewall to avoid venting gas in the direction of critical vehicle components or occupant arears). Features of the various example battery packs herein may generally be combined without limitation, absent express statements herein to the contrary.

The foregoing is merely illustrative of the principles of this disclosure, and various modifications may be made by those skilled in the art without departing from the scope of this disclosure. The above-described embodiments are presented for purposes of illustration and not of limitation. The present disclosure also can take many forms other than those explicitly described herein. Accordingly, it is emphasized that this disclosure is not limited to the explicitly disclosed methods, systems, and apparatuses, but is intended to include variations to and modifications thereof, which are within the spirit of the following paragraphs.

While some portions of this disclosure may refer to examples, any such reference is merely to provide context to the instant disclosure and does not form any admission as to what constitutes the state of the art.

What is claimed is:

1. A battery pack for an electric vehicle, comprising:

a pack housing comprising a crossmember between a first bay and a second bay;

a first plurality of battery cells arranged within the first bay;

a second plurality of battery cells arranged within the second bay, wherein the crossmember comprises a multidirectional vent passage configured to allow venting of pressurized fluid from one or more of the first plurality of battery cells and the second plurality of battery cells to pass between the first bay and the second bay; and

a pressure release valve arranged in a sidewall of the pack housing and configured to release pressure based on the venting of the pressurized fluid between the first bay and the second bay.

2. The battery pack of claim **1**, wherein the multidirectional vent passage is configured to modify a first pressure with the first bay and a second pressure with the second bay.

3. The battery pack of claim **1**, wherein the pressure release valve is a one-way pressure release valve to release pressure based at least on the venting through the multidirectional vent passage.

4. The battery pack of claim **1**, further comprising a plug pressure release valve and a deformable pressure release valve, wherein the plug pressure release valve is configured to permit the venting with the first and second bays in response to a difference between a first internal pack pressure and an external pressure.

5. The battery pack of claim **4**, wherein:

the difference between the first internal pack pressure and the external pressure is a first pressure differential;

the plug pressure release valve is configured to permit venting in response to being exposed to the first pressure differential;

the deformable pressure release valve is configured to permit venting in response to being exposed to a second pressure differential greater than the first pressure differential; and

the pressure release valve is configured to permit venting in response to a third pressure differential that is less than the second pressure differential and greater than the first pressure differential.

6. The battery pack of claim **1**, further comprising a deformable pressure release valve configured to permit the venting with the first and second bays in response to an internal pack pressure meeting a threshold pressure.

7. The battery pack of claim **1**, further comprising a deformable pressure release valve, wherein the deformable pressure release valve comprises a structure configured to deform in response to at least one of pressurized or heated fluid within the battery pack.

8. The battery pack of claim **1**, wherein the pressure release valve comprises an actuatable membrane pressure release valve structure, an umbrella valve or combinations thereof.

9. The battery pack of claim **1**, wherein the pressure release valve comprises at least one opening covered by a membrane, a moveable cover, or a rotating enclosure surface, or combinations thereof, that is configured to release or equalize pressure when arranged such that the at least one opening is uncovered.

10. The battery pack of claim **1**, wherein:

the multidirectional vent passage is positioned at a lower portion of the first and second bays above a floor surface of the battery pack;

the multidirectional vent passage has a width at least twice as great as a height of the multidirectional vent passage;

the lower portion of the first and second bays comprises a lower portion of the crossmember defining the first and second bays; and at least one of the first and second bays comprises at least one crossmember comprised of a second multidirectional vent passage.

11. The battery pack of claim **1**, wherein the battery pack includes at least two crossmembers defining in part at least three bays, each crossmember of the at least two crossmembers comprising respective multidirectional vent passages.

12. The battery pack of claim **1**, wherein the first and second pluralities of battery cells are raised above a floor structure of the battery pack, thereby allowing fluid flow within each bay along the floor structure beneath the first and second pluralities of battery cells.

13. The battery pack of claim **12**, wherein allowing fluid flow comprises enabling unrestricted trajectories of fluid to propagate between bays through multidirectional vent passages.

14. The battery pack of claim **1**, wherein at least one single membrane venting structure is embedded in a first end of the battery pack and at least one dual membrane venting structure is embedded in a second end of the battery pack.

15. A battery pack for an electric vehicle, comprising:

a pack enclosure comprising a crossmember that at least partially forms first and second bays that are adjacent;

a first plurality of battery cells arranged within the first bay;

a second plurality of battery cells arranged within the second bay, wherein the crossmember comprises a multidirectional vent passage configured to modify pressure based on venting of pressurized fluid from one or more of the first and second pluralities of battery cells to pass between the first bay and the second bay; and

a one-way pressure release valve positioned in a sidewall of the pack enclosure, wherein the one-way pressure release valve is configured to release pressure based on the venting of the pressurized fluid between the first bay and the second bay.

16. The battery pack of claim **15**, further comprising a plug pressure release valve and a deformable pressure release valve, wherein;

the plug pressure release valve is configured to permit the venting of the pressurized fluid in response to a first internal pack pressure below a first threshold pressure; and

the deformable pressure release valve is configured to permit the venting of the pressurized fluid in response to a second internal pack pressure exceeding a second threshold pressure greater than the first threshold pressure.

17. The battery pack of claim **15**, wherein the one-way pressure release valve comprises at least one opening covered by a membrane, a moveable cover, or a rotating enclosure surface or combinations thereof that is configured to release or equalize pressure when arranged such that the at least one opening is uncovered.

18. A method for venting a battery pack, the method comprising:

constructing a plurality of bays each with a respective crossmember using at least two sidewalls, wherein:

each respective crossmember comprises a multidirectional vent passage,

a first plurality of battery cells is arranged within a first bay of the plurality of bays,

a second plurality of battery cells is arranged within a second bay of the plurality of bays, and

the multidirectional vent passage between the first bay and the second bay is configured to allow venting of a pressurized fluid from one or more of the first plurality of battery cells and the second plurality of battery cells to pass between the first bay and the second bay; and

providing a pressure release valve with a pack housing such that pressure is released based at least on the multidirectional vent passage allowing venting of the pressurized fluid towards the pressure release valve.

19. The method of claim **18**, further comprising:

forming the plurality of bays by arranging at least a first wall and a second wall such that a top edge of the first wall and the second wall are laterally aligned or parallel, wherein a first end of the crossmember is fixedly attached to the first wall and a second end of the crossmember, opposite the first end, is fixedly attached to the second wall;

arranging the first plurality of battery cells in the first bay; and

arranging the second plurality of battery cells in the second bay.

20. The method of claim **19**, further comprising:

forming at least one first guiding rib in an extruded material comprising the first wall wherein the at least one first guiding rib connects an opening on an inner surface of the first wall to an opening on an outer surface of the first wall;

forming at least one second guiding rib in an extruded material comprising the second wall wherein the at least one second guiding rib connects an opening on an inner surface of the second wall to an opening on an outer surface of the second wall; and

machining an opening in each of the outer surface of the first wall and the outer surface of the second wall, wherein each of the openings comprises a corresponding feature configured to receive at least one mounting extension of the pressure release valve.

* * * * *